Practical 8

AIM : Performing Advanced Analytics using Trends,Clustering, and Forecasting
a) Add and interpret trend lines (linear, exponential)
b) Customize and analyze trend models
c) Perform clustering on datasets
d) Analyze distributions using built-in tools
e) Apply forecasting techniques
f) Compare different statistical models
g) Export and interpret statistical results

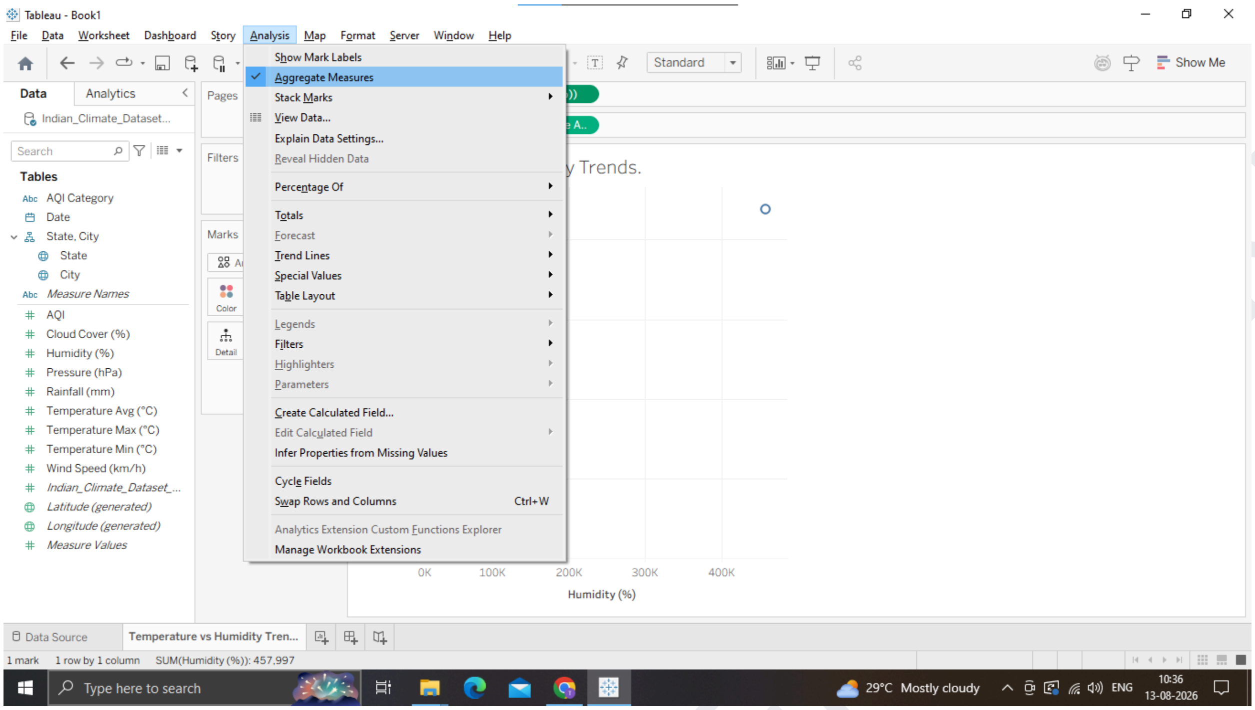
Dataset: Indian Climate Dataset (2024–2025)

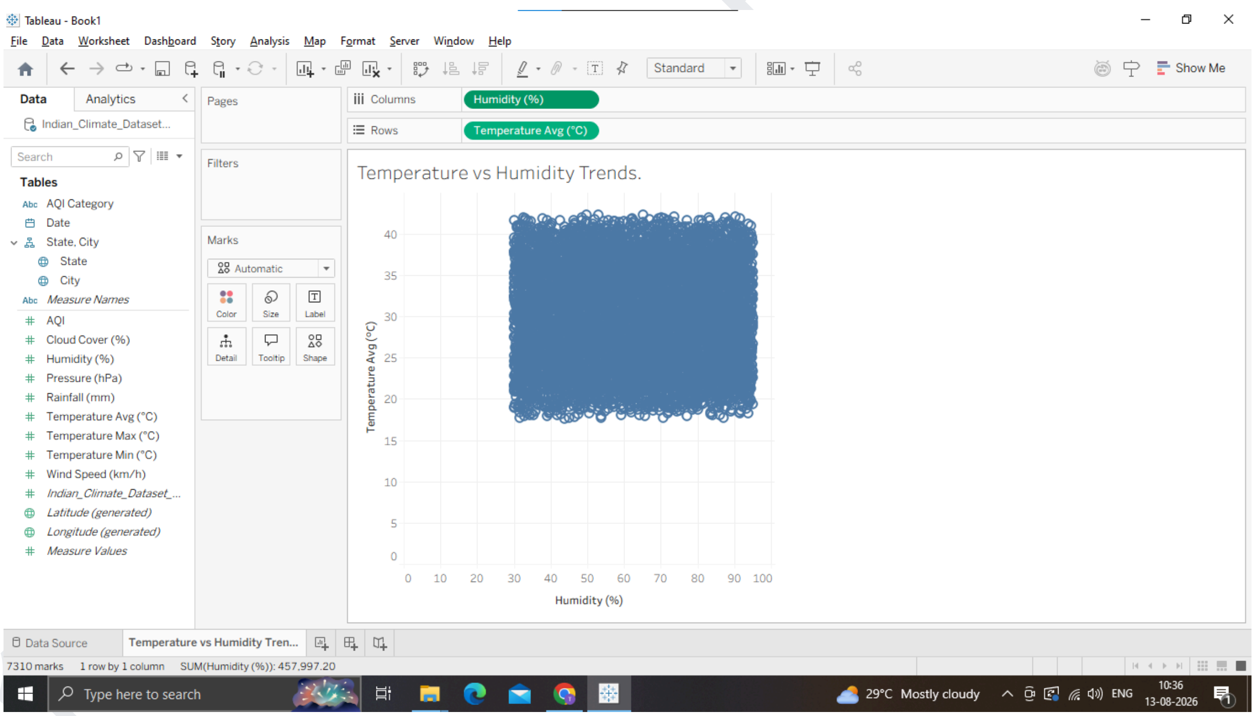
Kaggle link: https://www.kaggle.com/datasets/ankushnarwade/indian-climate-dataset20242025

A: Add and Interpret Trend Lines (Linear, Exponential)

1. Connect Dataset & Build a Scatter Plot / Line Chart

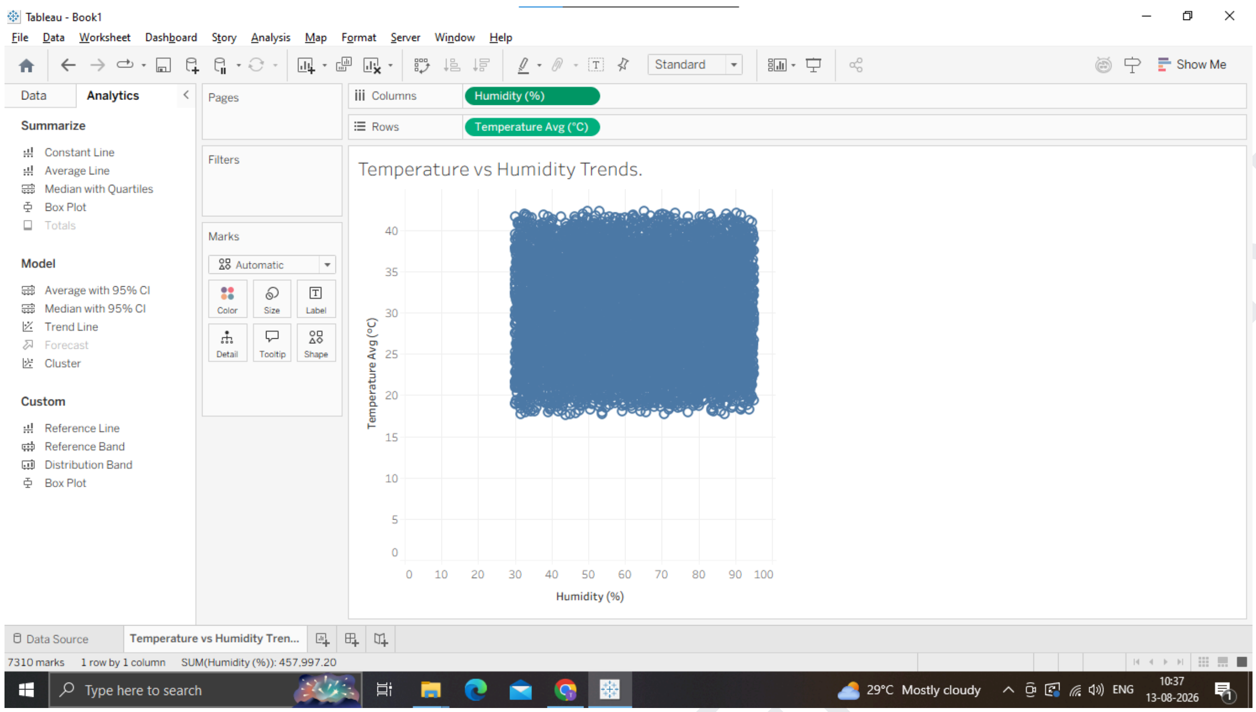
1. Open Tableau Desktop and load Indian_Climate_Dataset_2024_2025.csv.
2. Open a new worksheet and rename it to Temperature vs Humidity Trends.
3. Drag Humidity (%) to the Columns shelf.
4. Drag Temperature_Avg (°C) to the Rows shelf.
5. In the top menu, go to Analysis and uncheck Aggregate Measures (so each data point is plotted individually rather than summed up).



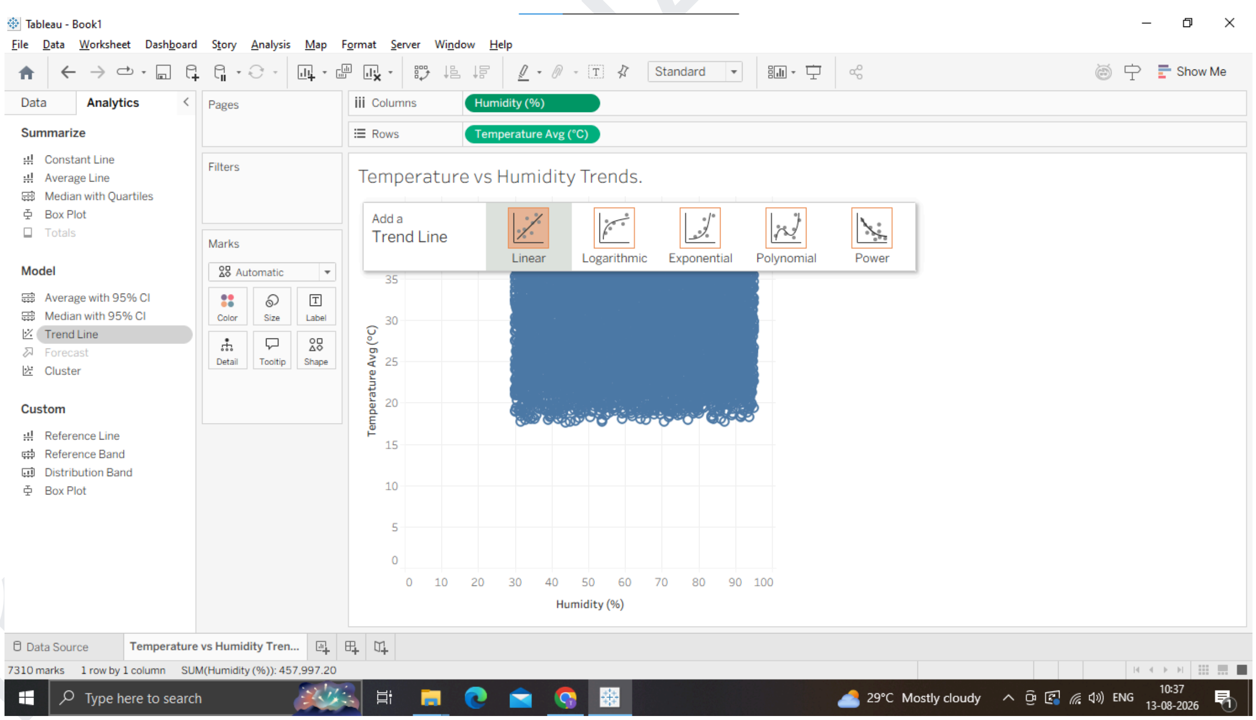


2. Add a Linear Trend Line

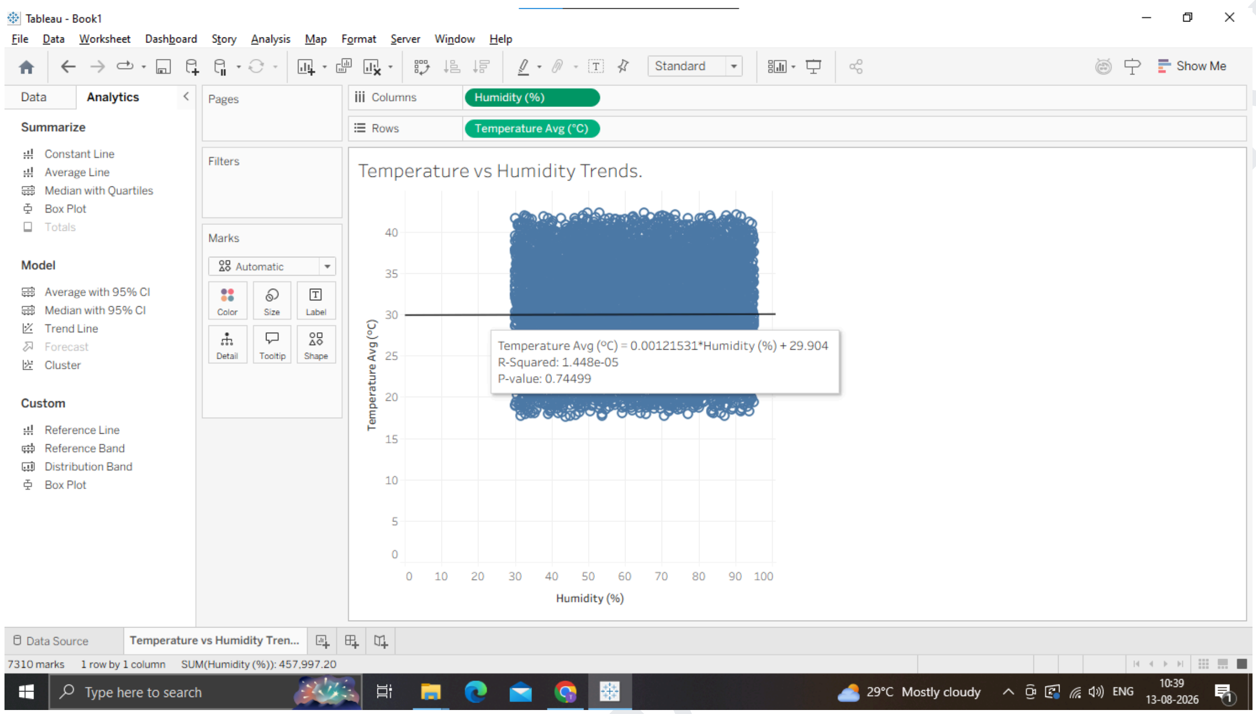
1. Switch to the Analytics pane on the left sidebar (tab next to Data).



2. Under Model, drag Trend Line onto the canvas and drop it onto Linear.

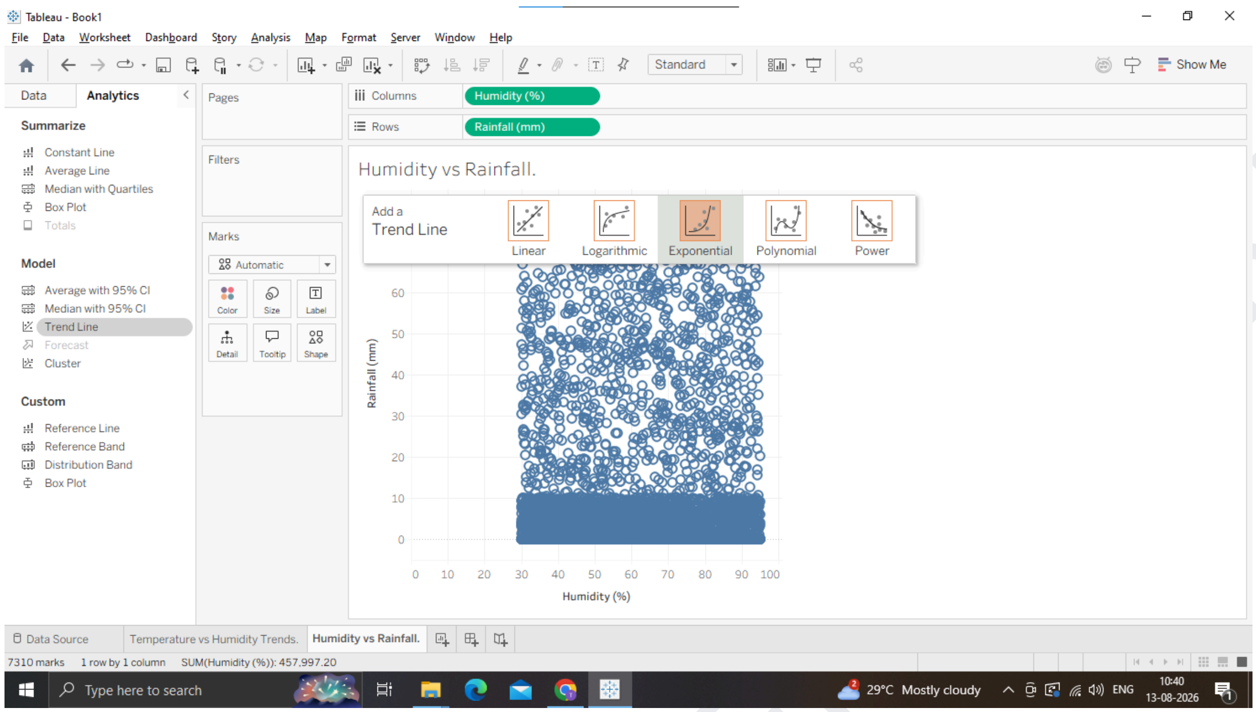


3. Hover over the trend line to observe the statistical popup:
   o It displays the linear equation: Temperature_Avg = a x Humidity + b
   o It displays R2 (coefficient of determination) and P-value (statistical significance).



3. Add an Exponential Trend Line for Non-Linear Measures (Rainfall vs. Humidity)

1. Open a new worksheet named Humidity vs Rainfall.
2. Drag Humidity (%) to Columns and Rainfall (mm) to Rows.
3. Uncheck Analysis Aggregate Measures.
4. From the Analytics pane, drag Trend Line onto Exponential (or right-click the canvas Trend Lines Show Trend Lines Edit Trend Lines and change the model to Exponential).



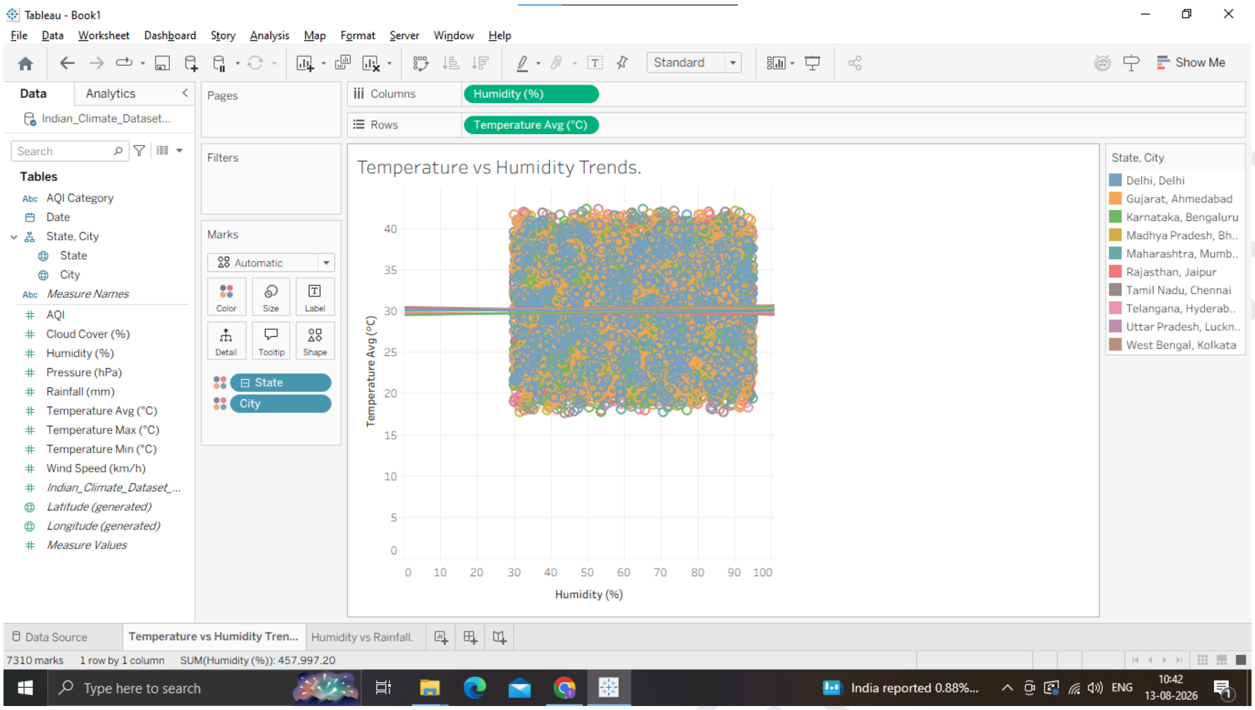
4. Interpret the Results

Linear Model: Describes direct proportional/inverse relationships (e.g., checking if higher humidity steadily decreases or increases average temperature).
Exponential Model: Captures accelerating relationships (e.g., how rainfall spikes exponentially once humidity crosses higher thresholds).

Why we are doing it
Identify Relationships: A single dot on a scatter plot tells us about one day's weather, but a trend line shows the underlying mathematical relationship across thousands of data points.
Quantify Correlation: It allows us to move from guessing ("It feels hotter when it's humid") to measuring ("For every 10% increase in humidity, average temperature drops by XoC").
Statistical Validation: The R2 value tells us how much variance is explained by the model, while the P-value proves whether the trend is statistically significant or just random noise.

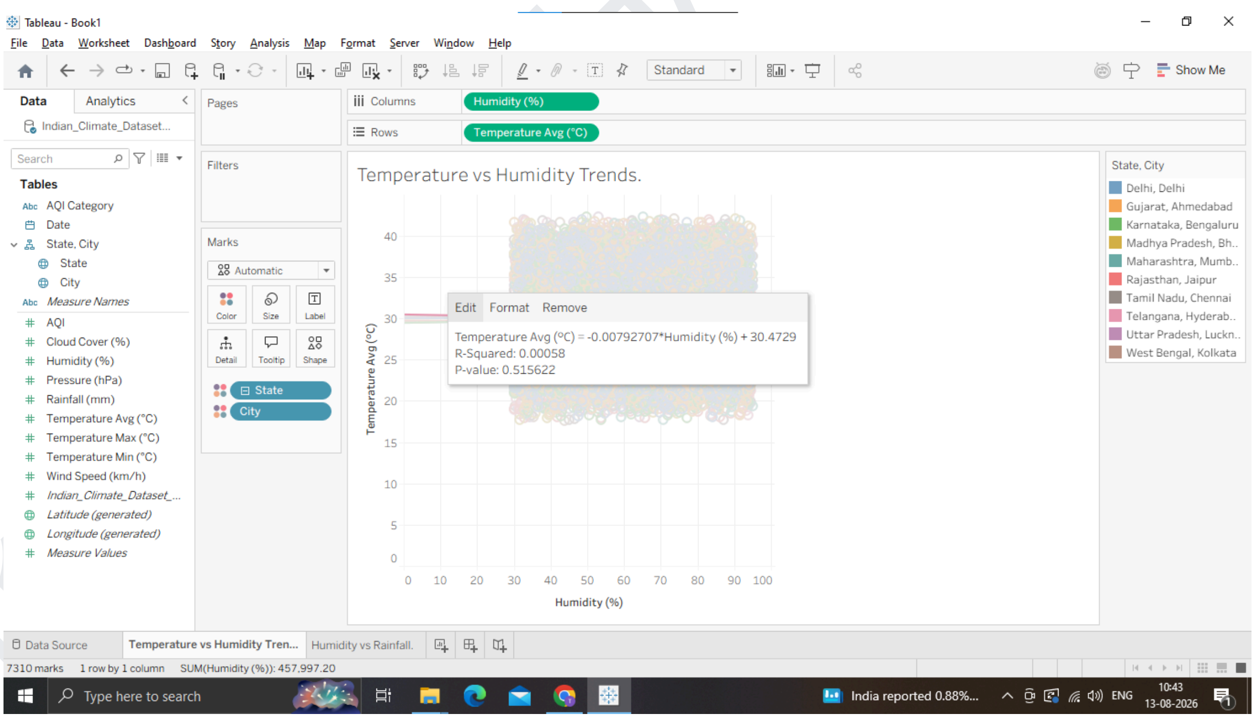
B: Customize and Analyze Trend Models

1. Disaggregate Trend Lines by City
1. Open your Temperature vs Humidity Trends sheet from Part A (or create a duplicate).
2. Drag City from the Data pane onto Color on the Marks card. o Notice what happens: Instead of one global trend line, Tableau automatically
computes 10 distinct trend lines, one for each city!
3. Observe how coastal cities (like Mumbai or Chennai) show different slopes compared to inland cities (like Delhi or Jaipur).



2 Compare Different Model Types (Polynomial & Logarithmic)

1. Right-click on any trend line in the view and select Edit Trend Lines…



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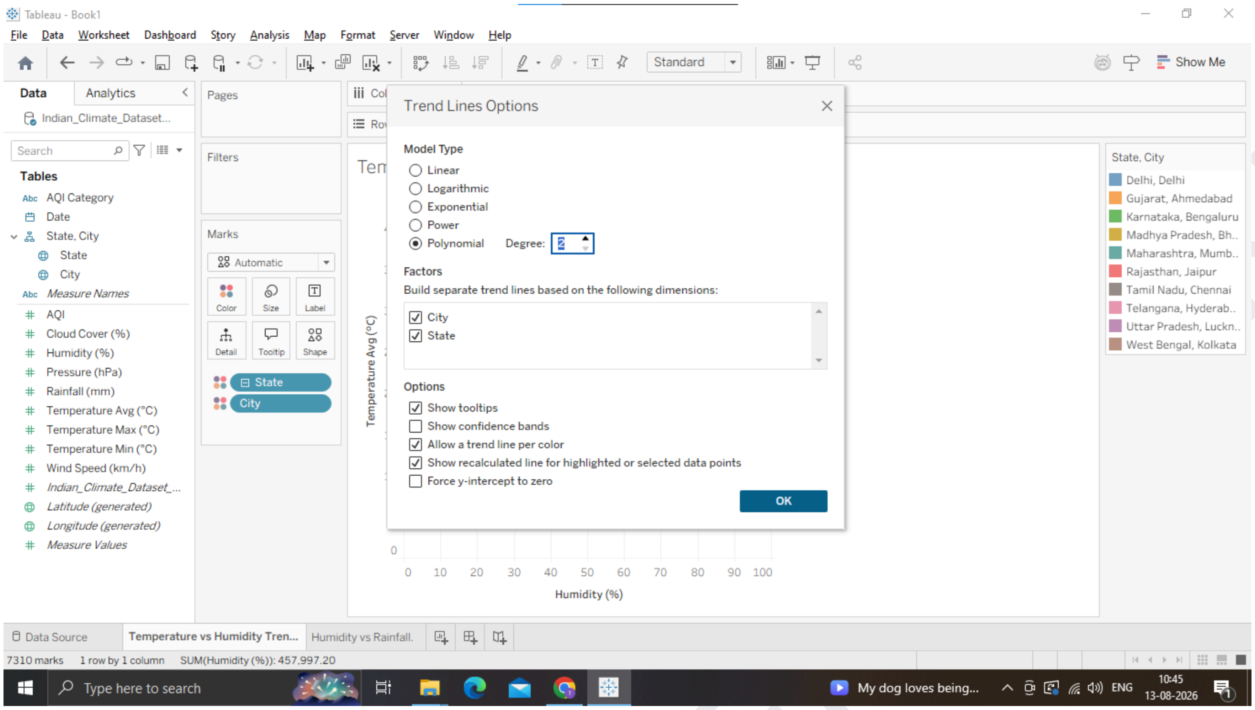
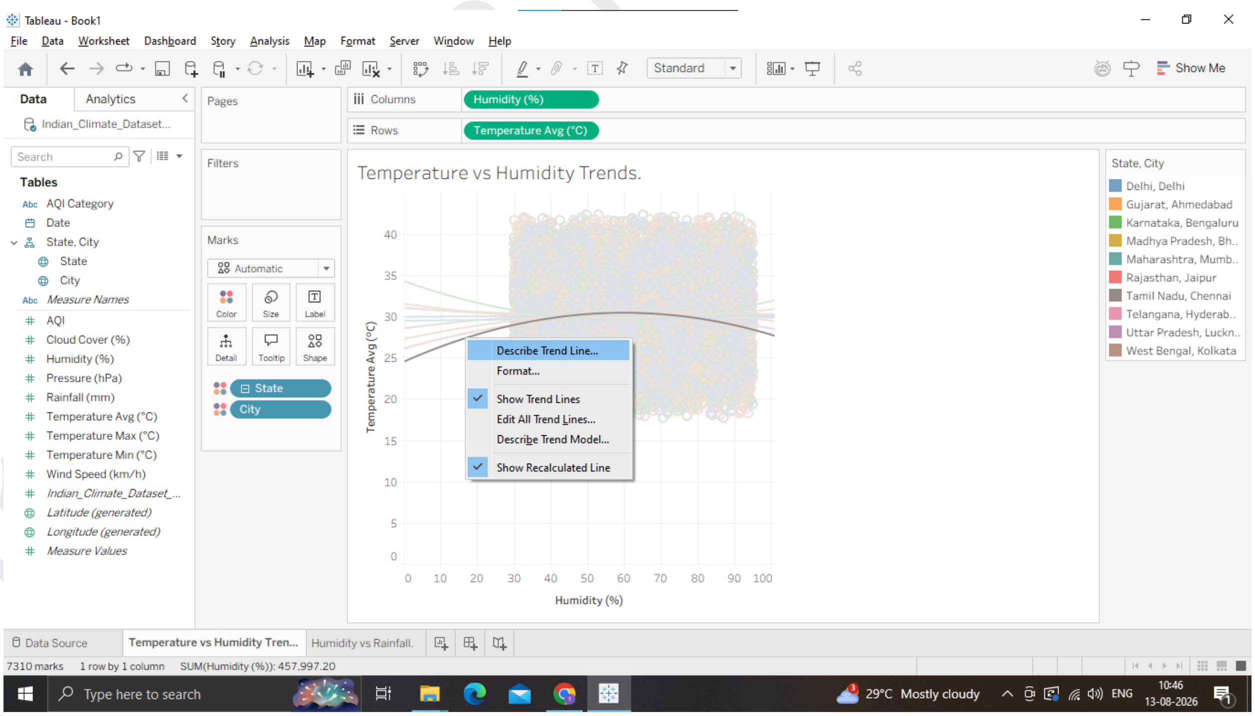
Data Visualization with Power BI and Tableau

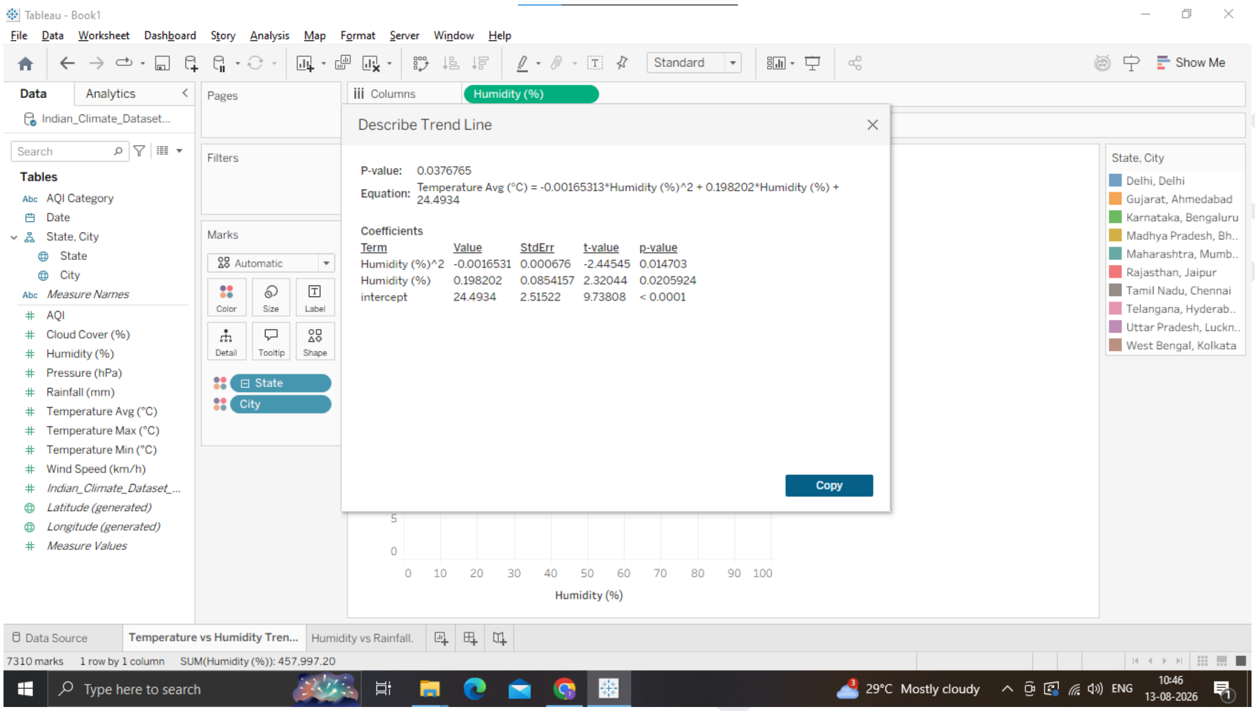


3. Access Full Statistical Summaries (Describe Model)

1. Right-click anywhere on the chart background or trend line and select Describe Trend Line...

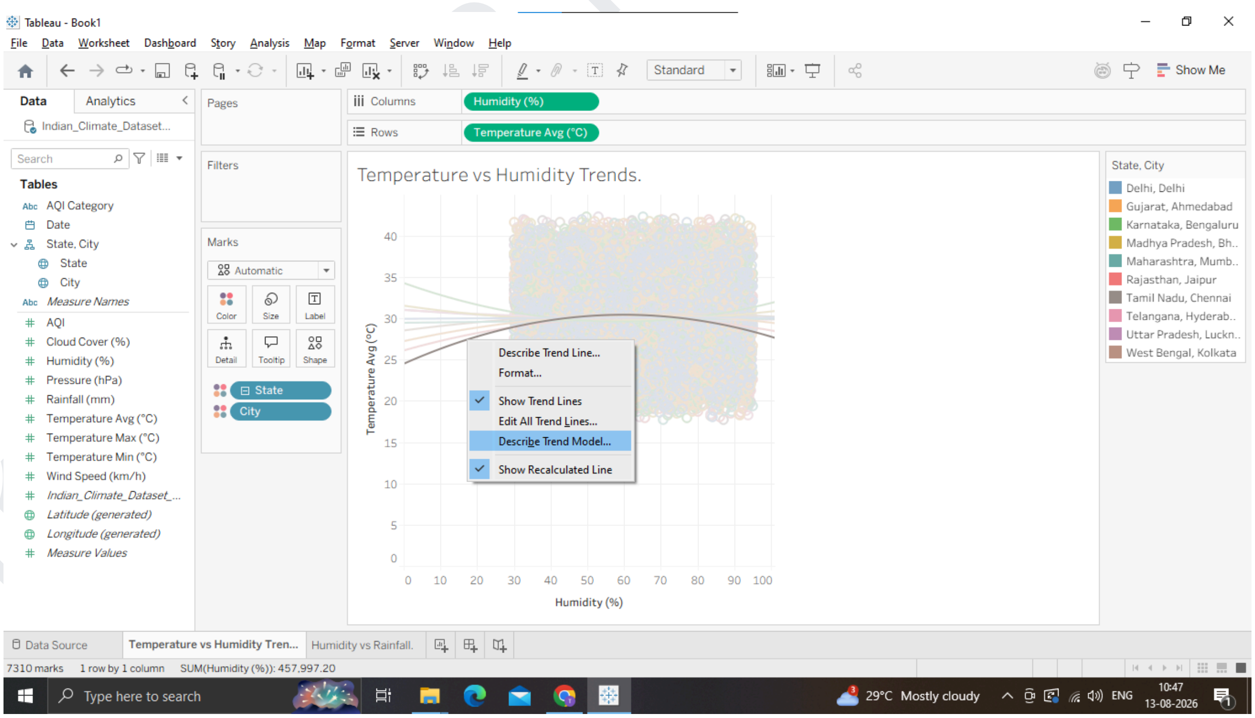
o This shows a quick overview of formula equations, R2 and P-value per city.

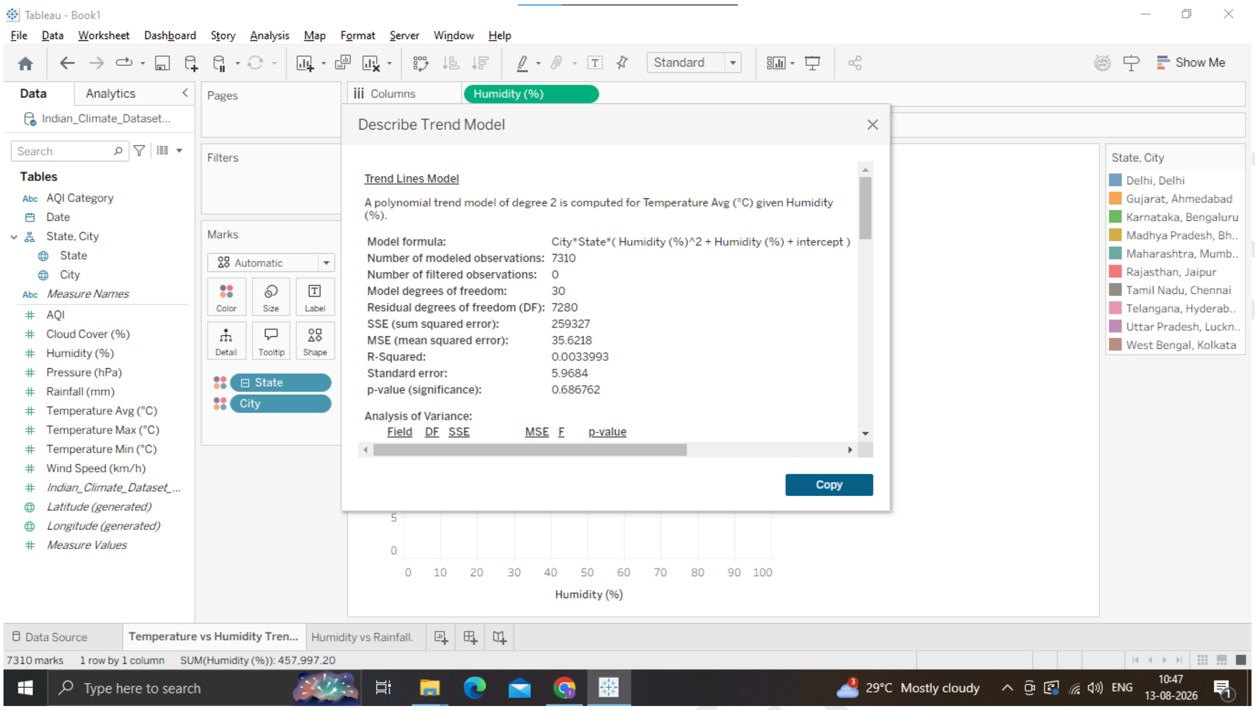




2. Next, right-click and select Describe Trend Model...

o A detailed window will pop up showing the ANOVA table, Degrees of Freedom, Sum of Squared Errors (SSE), and F-statistic.





3. Review these metrics to see which city's model explains the highest variance (R2)

Why we are doing it ?

Isolate Subgroups: In climate data, a single trend line across all 10 cities or 28 states can hide local climate realities. Customizing trend lines per category (e.g., per City or Region) lets us compare micro-climates directly.

Test Model Types: Comparing Linear, Logarithmic, Exponential, and Polynomial models allows us to find which mathematical formula best describes the physical phenomenon.

Deep Model Diagnostics: Going beyond the surface tooltip into Tableau's full Describe Trend Model window gives us detailed ANOVA (Analysis of Variance) tables, standard errors, and degrees of freedom required for rigorous reporting.

Usage

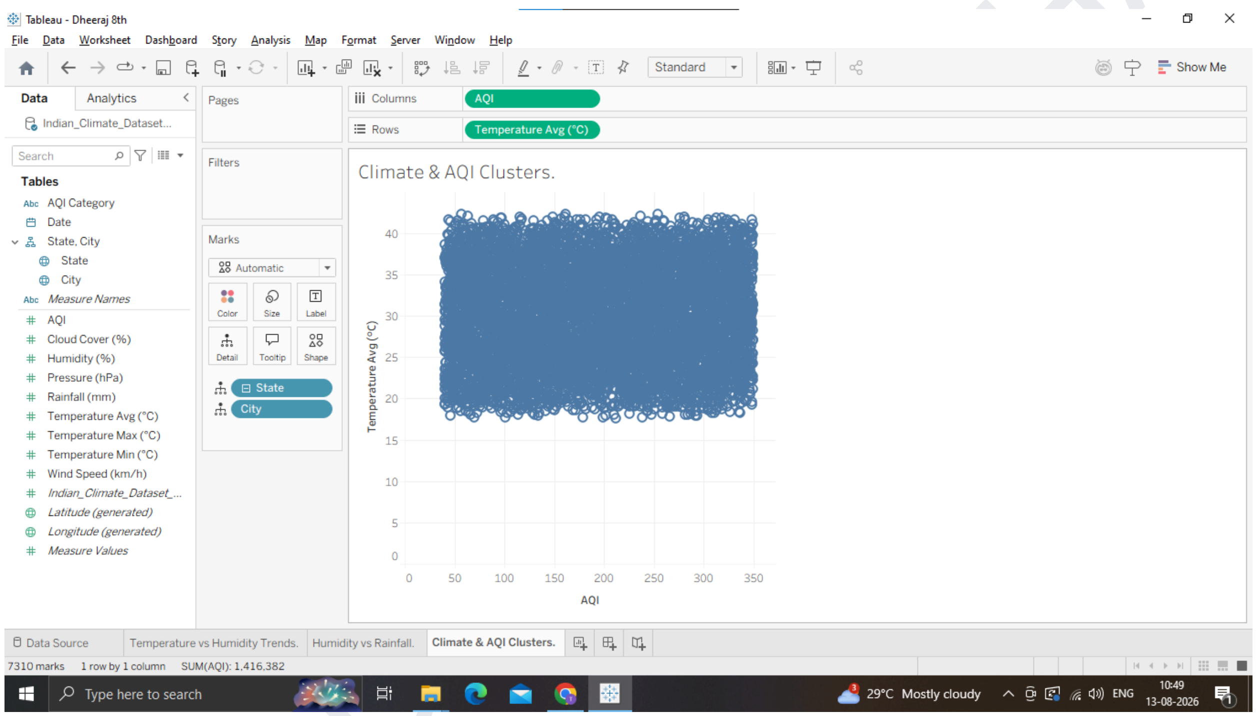
Regional Climate Policy: Disaggregating trends by city allows local governments (e.g., Delhi vs. Chennai) to design targeted air quality or heatwave action plans.

Smart Grid & HVAC Optimization: Utility companies use non-linear models per region to predict local electricity spikes during peak monsoon vs. summer months.

C: Perform Clustering on Datasets

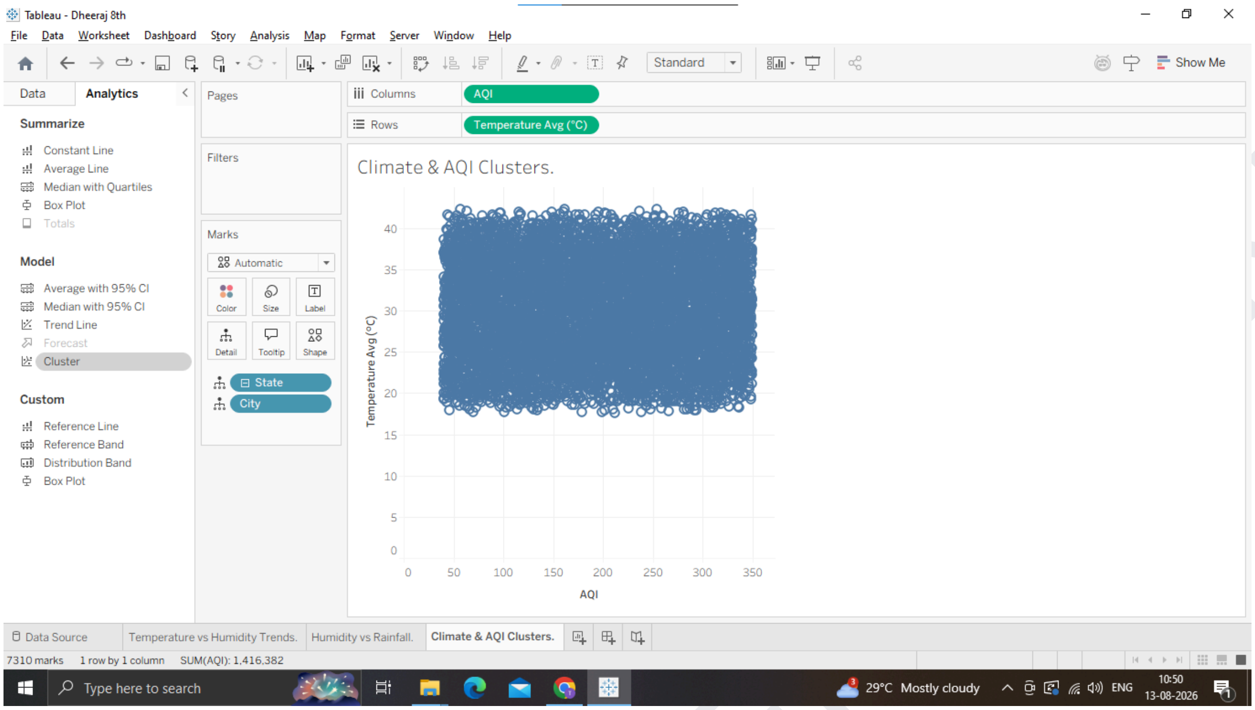
1. Build a Multi-Metric Scatter Plot Matrix

1. Open a new worksheet and name it Climate & AQI Clusters.
2. Drag AQI to Columns.
3. Drag Temperature_Avg (°C) to Rows.
4. In the top menu, go to Analysis and uncheck "Aggregate Measures".
5. Drag City to Detail on the Marks card (so points represent individual city-day observations).

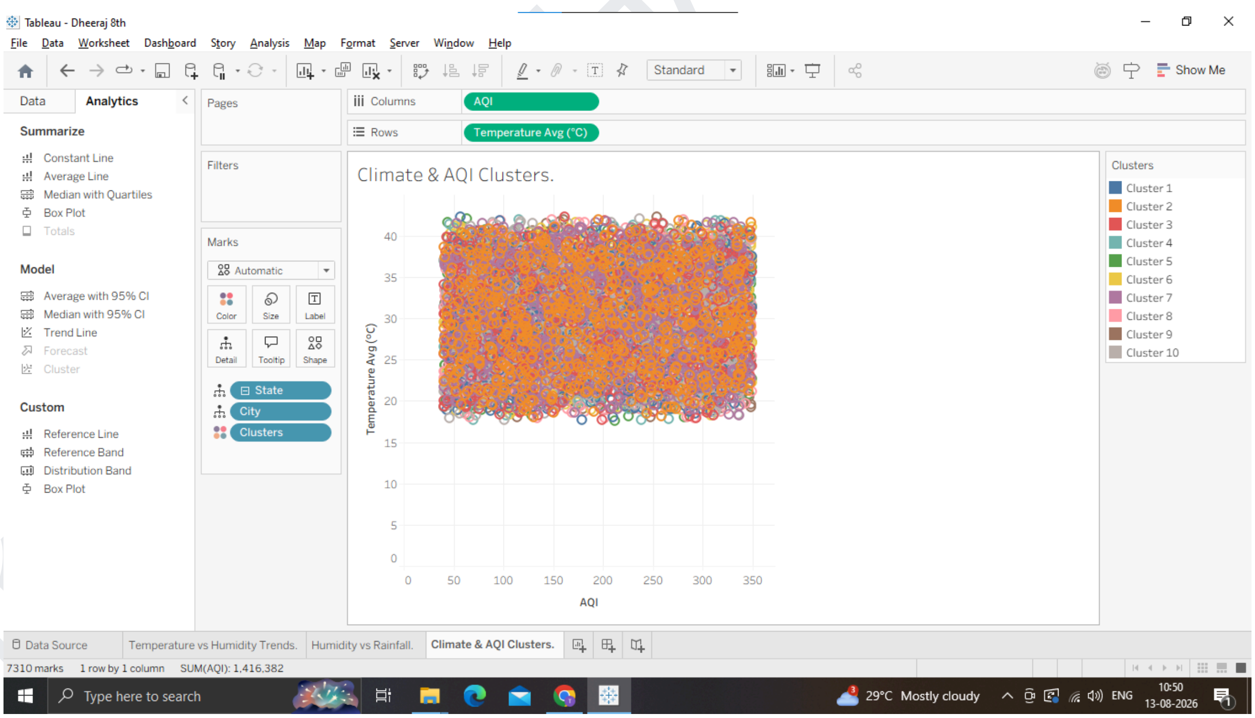


2. Apply Built-in Tableau Clustering

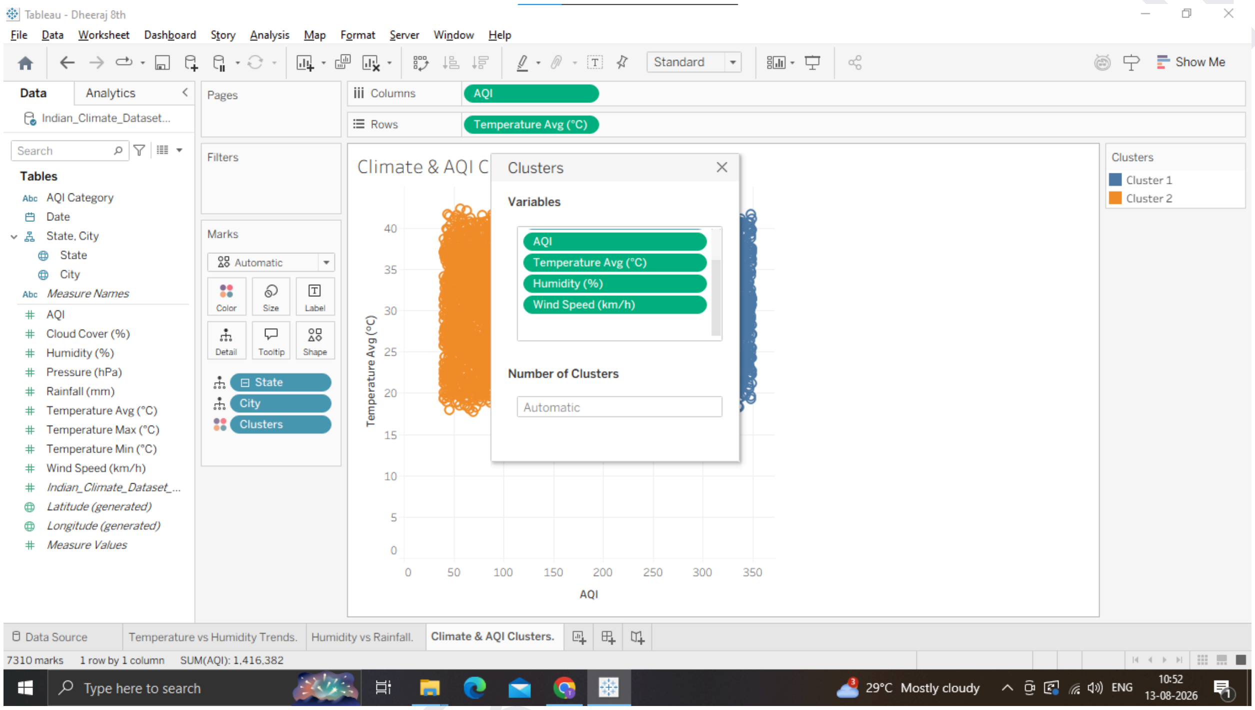
1. Switch to the Analytics pane on the left sidebar.
2. Under Model, drag Cluster directly onto the canvas.



3. A Clusters dialog box will appear on the right side of the view.



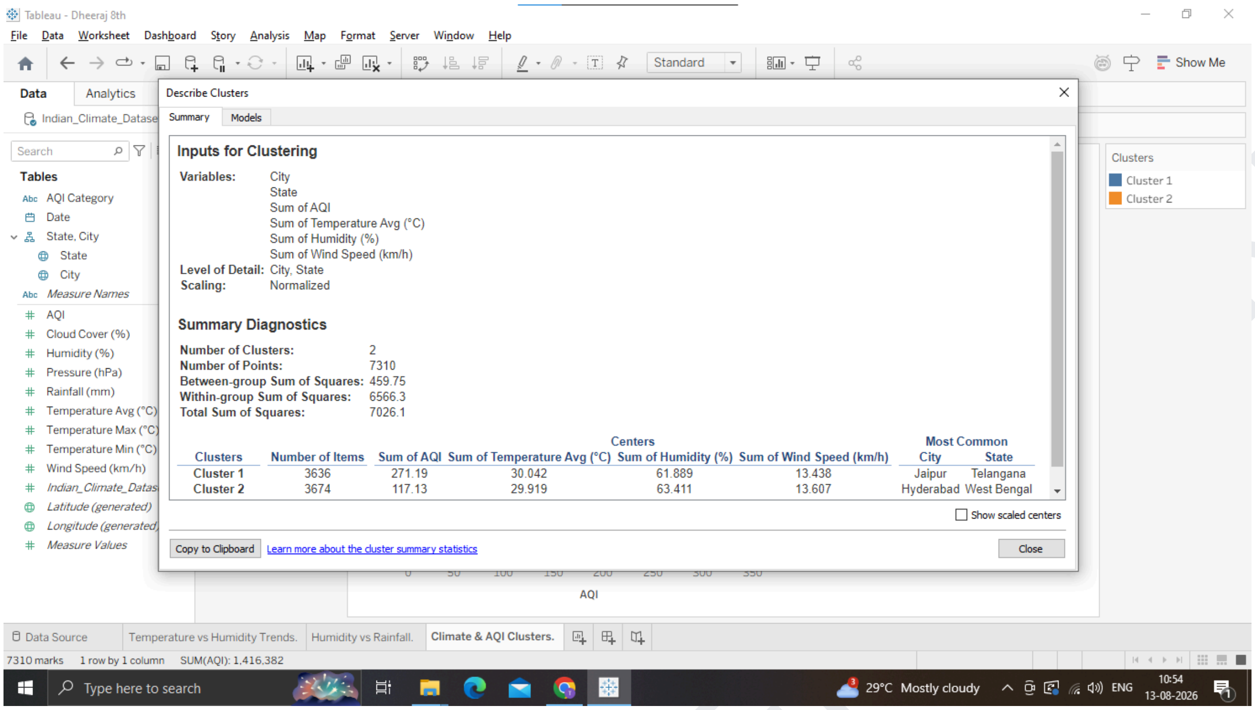
4. Drag additional variables into the Variables box to create a multi-variable cluster model (If you can't find Variable box Right Click Clusters in Marks Card,click on Edit Clusters and drag and drop the variables on the box):
o AQI
o Temperature_Avg (°C)
o Humidity (%)
o Wind_Speed (km/h)



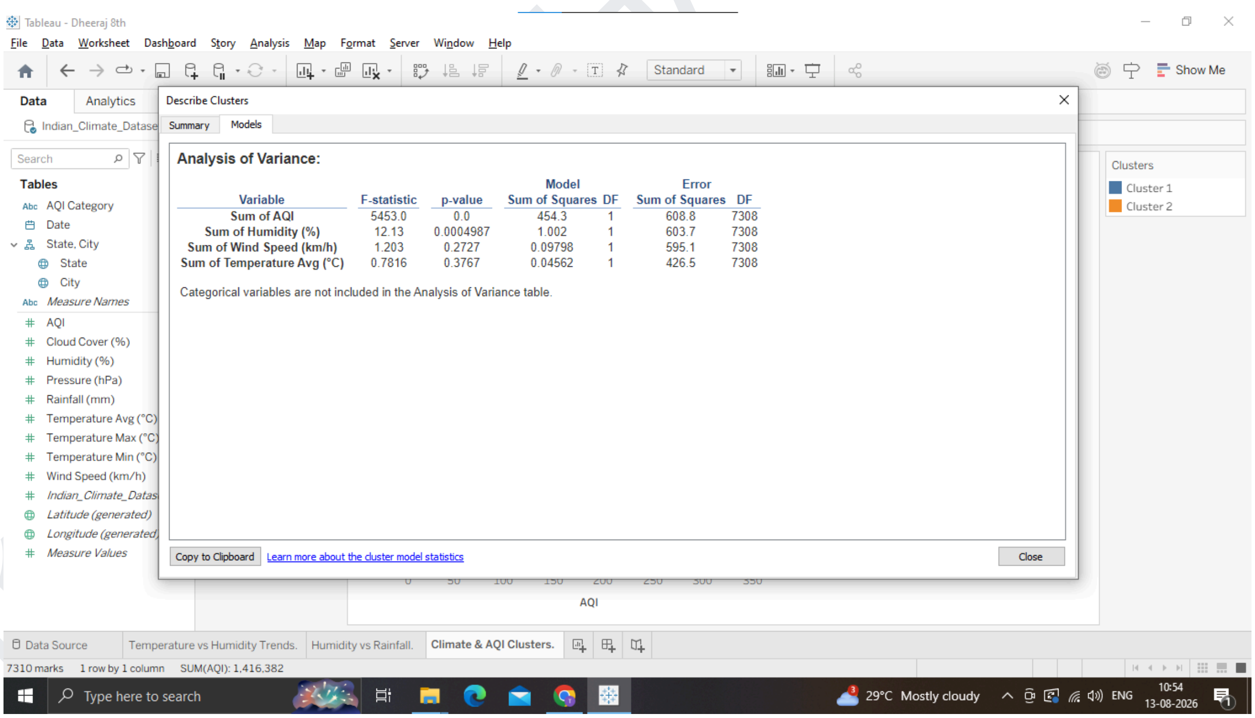
5. Under Number of Clusters, leave it as Automatic (Tableau will calculate the optimal number using K-means), or type 3 or 4 to explicitly create 3 or 4 distinct climate groups.
6. Close the cluster popup window. Tableau will color-code every data point by its assigned cluster (e.g., Cluster 1, Cluster 2, Cluster 3).

3. Analyze Cluster Profiles & Save as Dimension

1. Right-click on the Clusters field in the Marks card (or legend) and select Describe Clusters...
2. In the popup window, review two key tabs: o Summary: Shows the mean value of each metric (AQI, Temperature, Humidity, Wind Speed) for each cluster.



o Models: Shows ANOVA statistics (F-statistics and p-value) indicating which variables contributed most heavily to separating the clusters.



Why we are doing it ?

Automated Grouping: Instead of manually defining boundaries (e.g., "high AQI" or "low temperature"), clustering algorithms group similar data points together based on multiple variables simultaneously.

K-Means Algorithm: Tableau uses the K-Means clustering algorithm (using Euclidean distance) to discover hidden patterns, segmenting cities or days into distinct climate profiles.
Unsupervised Insight: It helps identify multidimensional weather states—like Hot & Dry, Cool & Monsoon, or Polluted & Humid—without needing predefined labels.

Usage

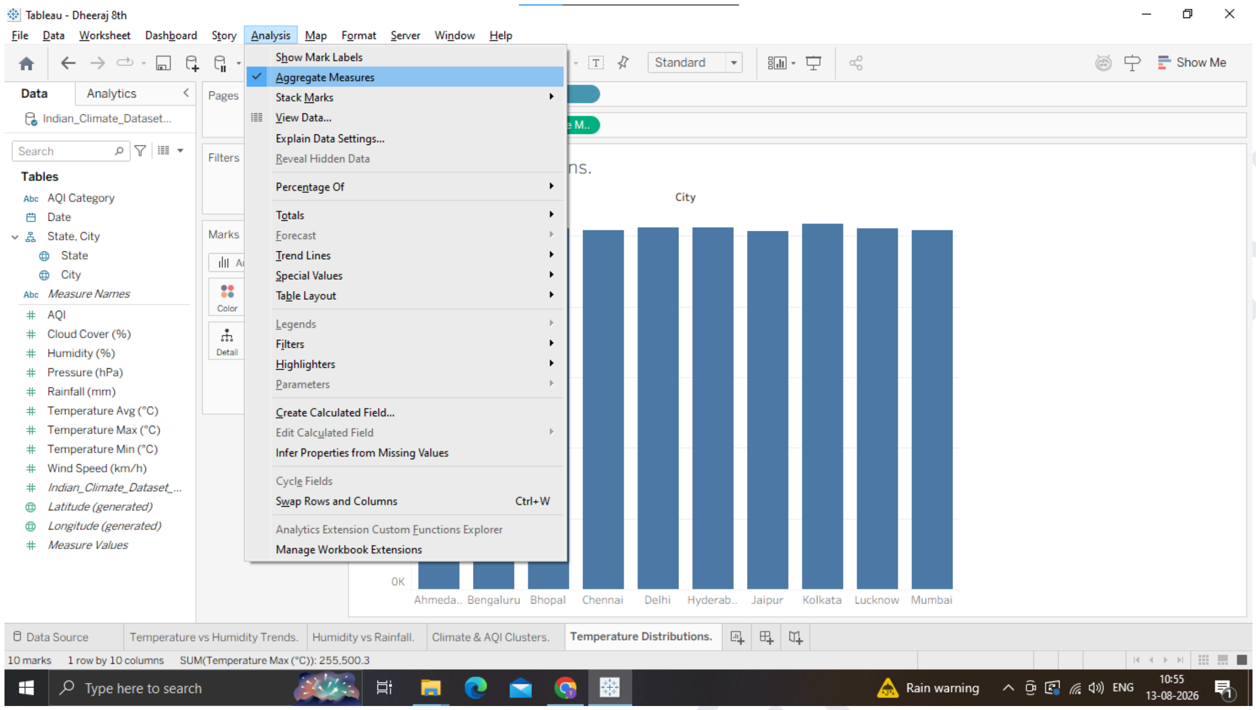
Air Quality & Health Hazards: Grouping days into clusters based on AQI, Temperature, and Wind Speed helps public health officials issue targeted pollution warnings.

Agricultural Zonation: Farmers and agronomists cluster regions with similar micro-climates (Rainfall, Humidity, Temperature) to select optimal crops and irrigation schemes.

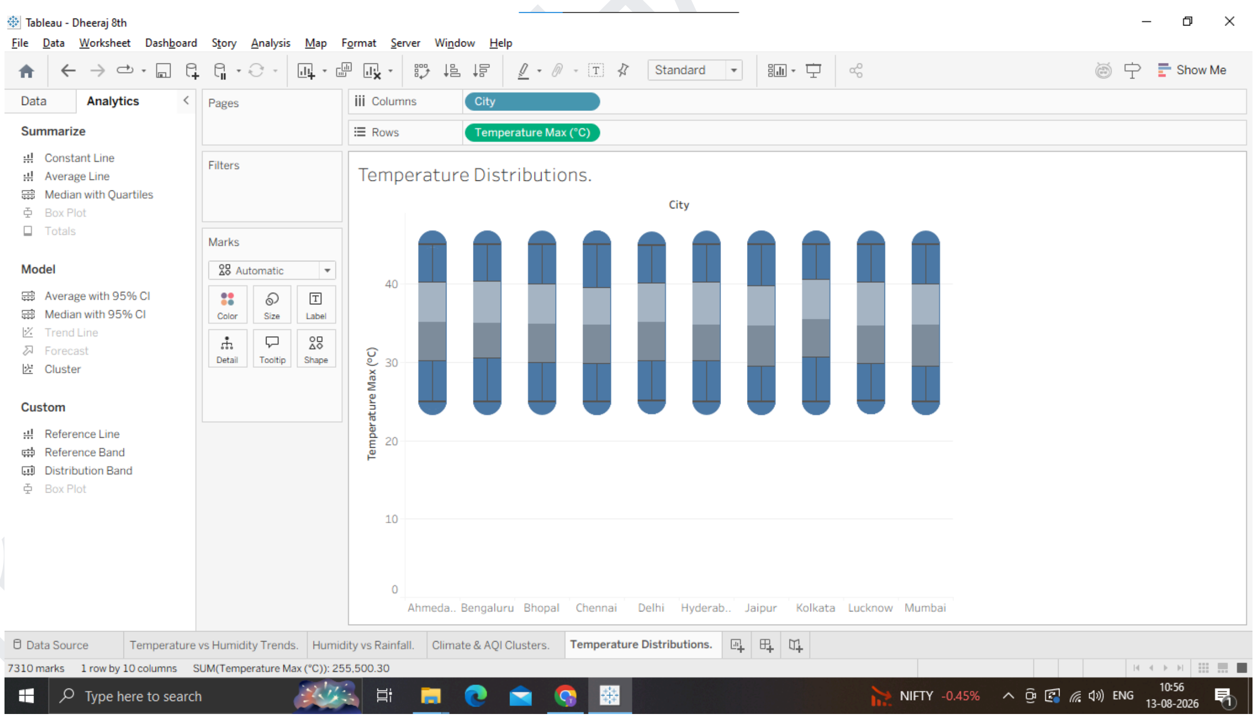
D: Analyze Distributions using Built-in Tools

1. Create a Box Plot for Temperature Distributions by City

1. Open a new worksheet and name it Temperature Distributions.
2. Drag City to Columns.
3. Drag Temperature_Max (°C) to Rows.
4. Tableau will default to SUM(Temperature_Max). Right-click the Temperature_Maxpill on Rows Measure (Sum) change it to Dimension (or go to Analysis menu and uncheck "Aggregate Measures").



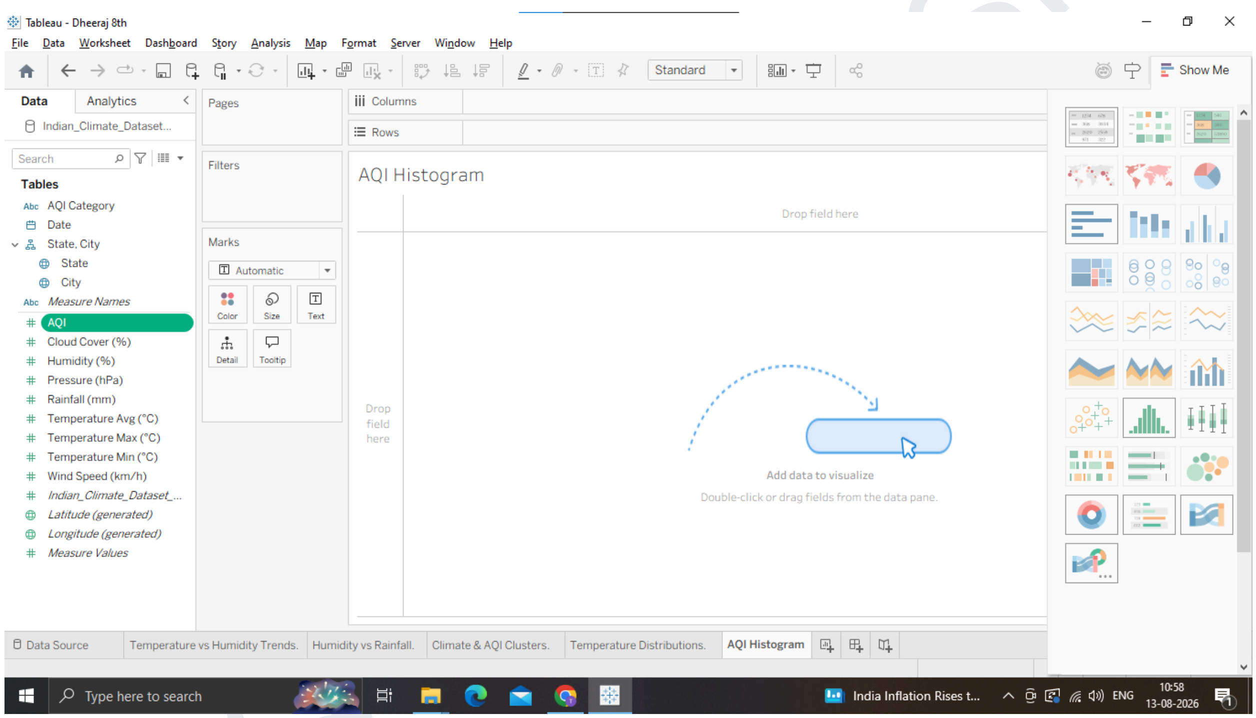
5. Switch to the Analytics pane on the left sidebar.
6. Under Custom, drag Box Plot onto the canvas and drop it on the chart.



7. Hover over any box plot to inspect key statistical metrics:
o Upper Whisker / Maximum: Upper fence boundary.
o Upper Hinge (Q3): 75th percentile.
o Median (Q2): 50th percentile.
o Lower Hinge (Q1): 25th percentile.
o Outliers: Individual points plotted beyond the whiskers.

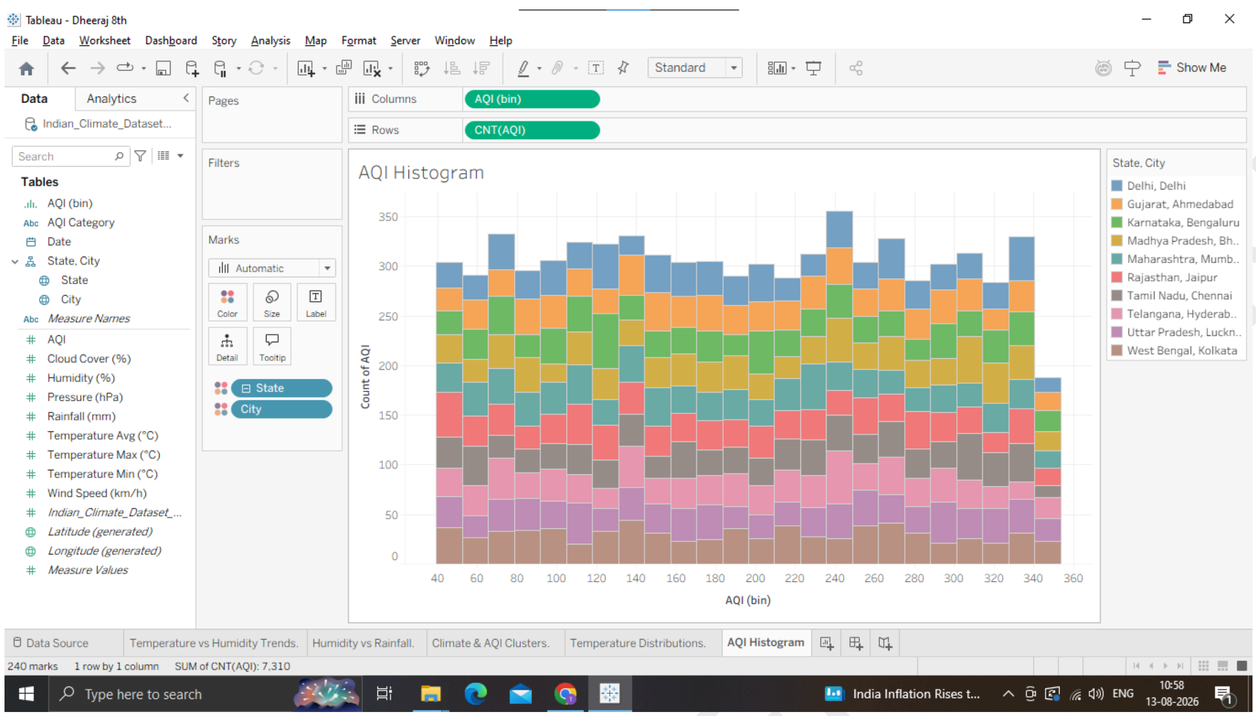
2. Build an AQI Histogram & Quantile/Percentile View

1. Open a new worksheet named AQI Histogram.
2. From the Data Pane, click on AQI.
3. Open the Show Me panel in the top-right corner of Tableau.



4. Select the Histogram icon.
(Tableau will automatically create an AQI (bin) dimension and place AQI (bin) on Columns and CNT(AQI) on Rows).

5. Drag City to Color on the Marks card to see how air quality distributions stack across cities.



Why we are doing it

Understand Data Spread: Averages (means) can be misleading, a city might have an average temperature of 28oC, but fluctuate wildly between 15oC and 41oC.

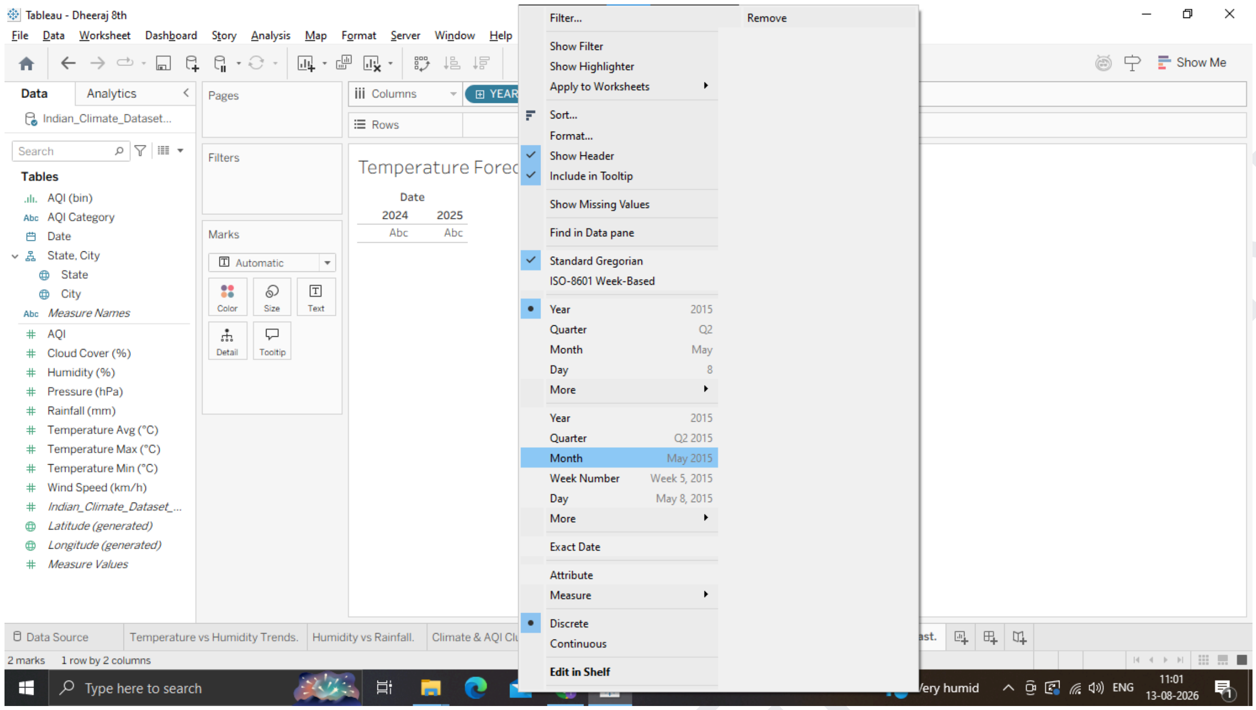
Detect Outliers & Skewness: Distribution tools like Box Plots and Histograms expose extreme events (e.g., severe pollution days or flash rainfall) and show whether data is normally distributed or heavily skewed.

Statistical Comparison: Box plots allow us to visually compare median values, interquartile ranges (IQR), and outlier thresholds across multiple cities side-by-side.

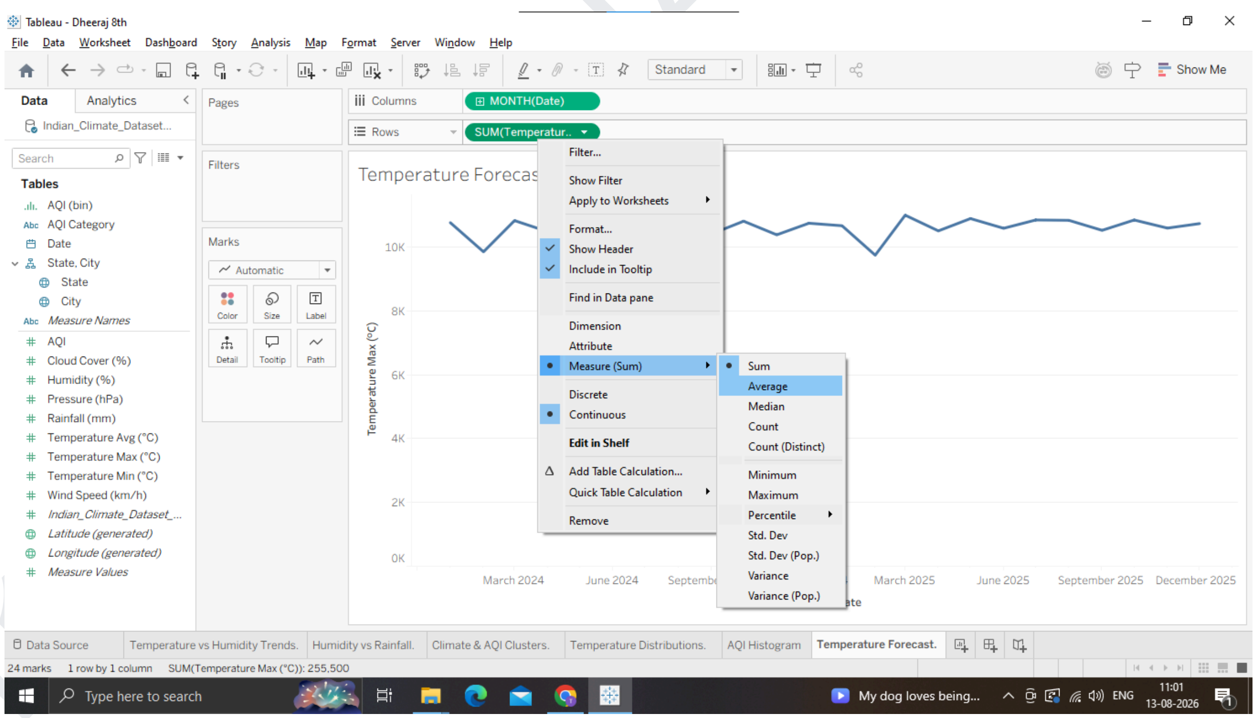
E: Apply Forecasting Techniques

1. Set Up a Time Series Line Chart

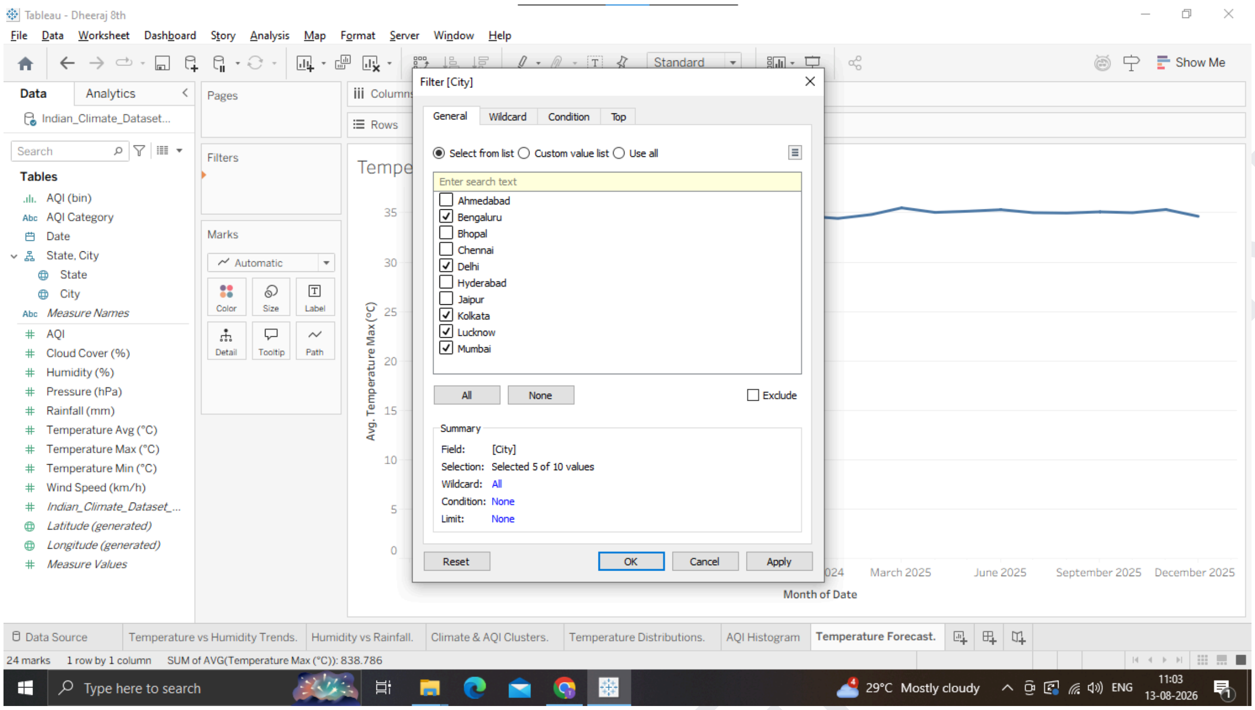
1. Open a new worksheet and name it Temperature Forecast.
2. Drag Date to Columns.
 o Note: Tableau will default to YEAR(Date). Click the dropdown arrow on the Datepill and change it to continuous Month (the second Month option in the menu,
 which displays as Month Year, e.g., May 2024).



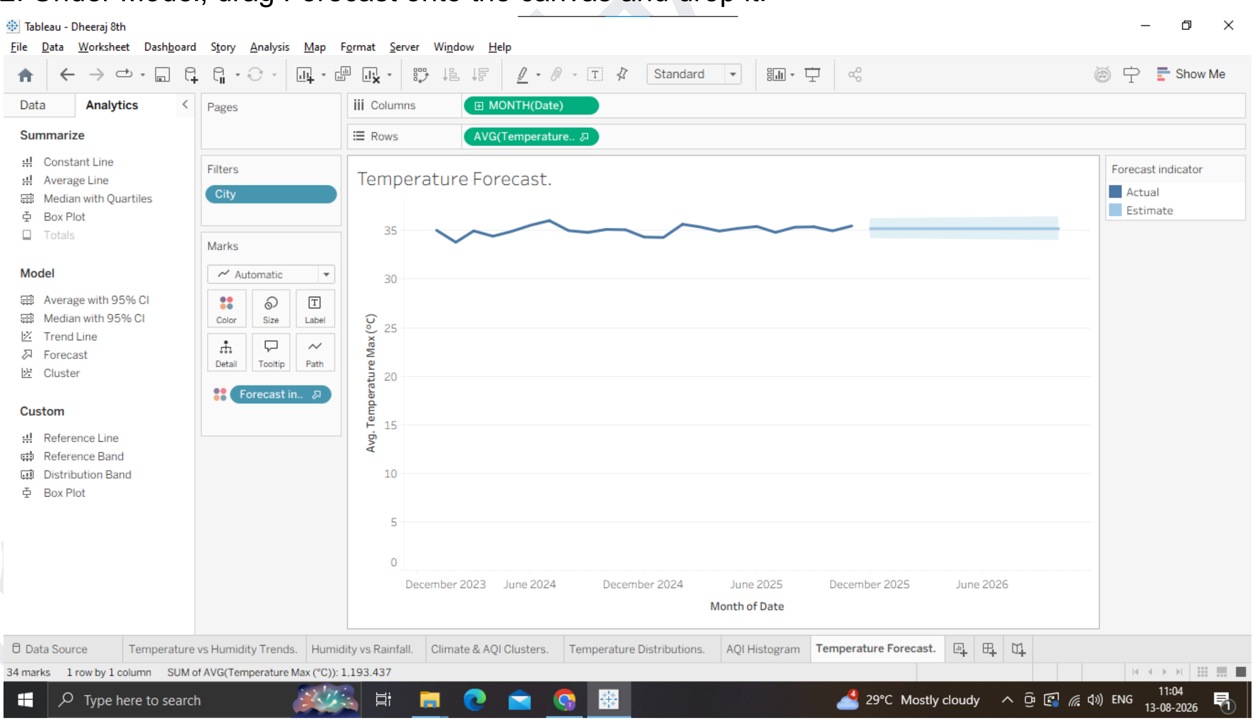
3. Drag Temperature_Max (°C) to Rows (ensure it is set to AVG(Temperature_Max)).



4. Drag City to Filters and select 2 or 3 cities (e.g., Mumbai, Delhi, Bengaluru) to keep the view clean.



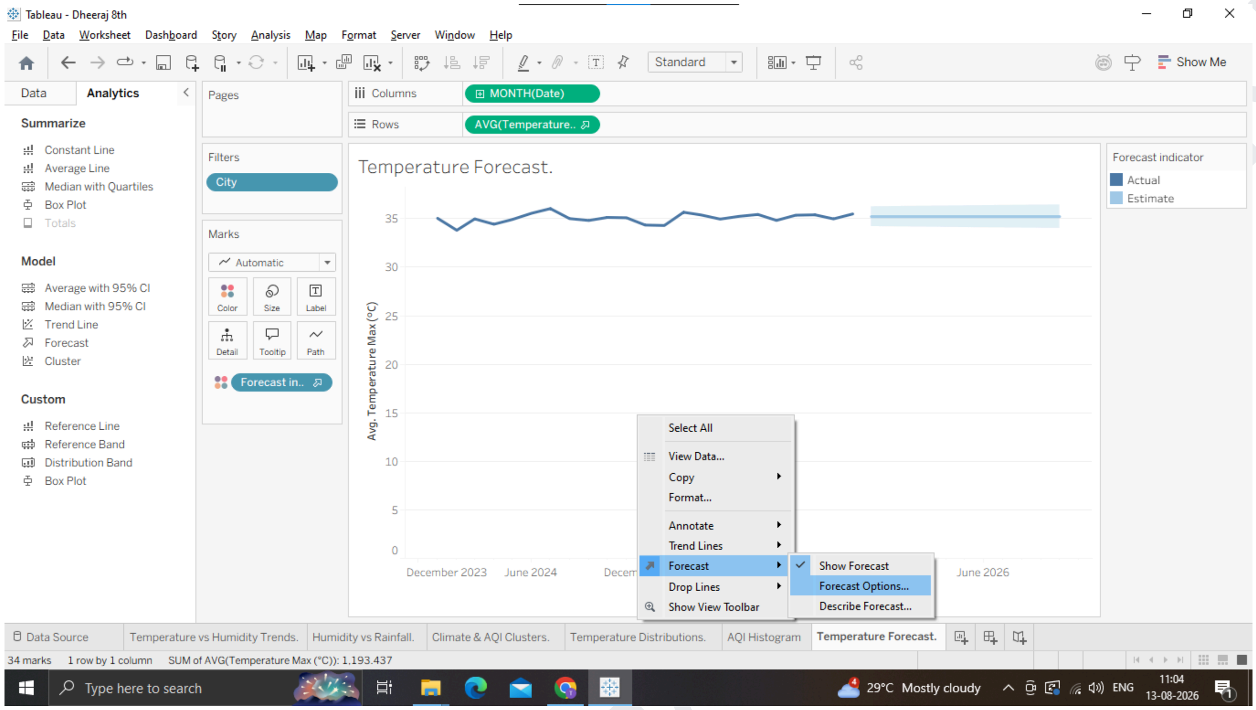
2. Enable Tableau Forecasting
1. Switch to the Analytics Pane on the left sidebar.
2. Under Model, drag Forecast onto the canvas and drop it.



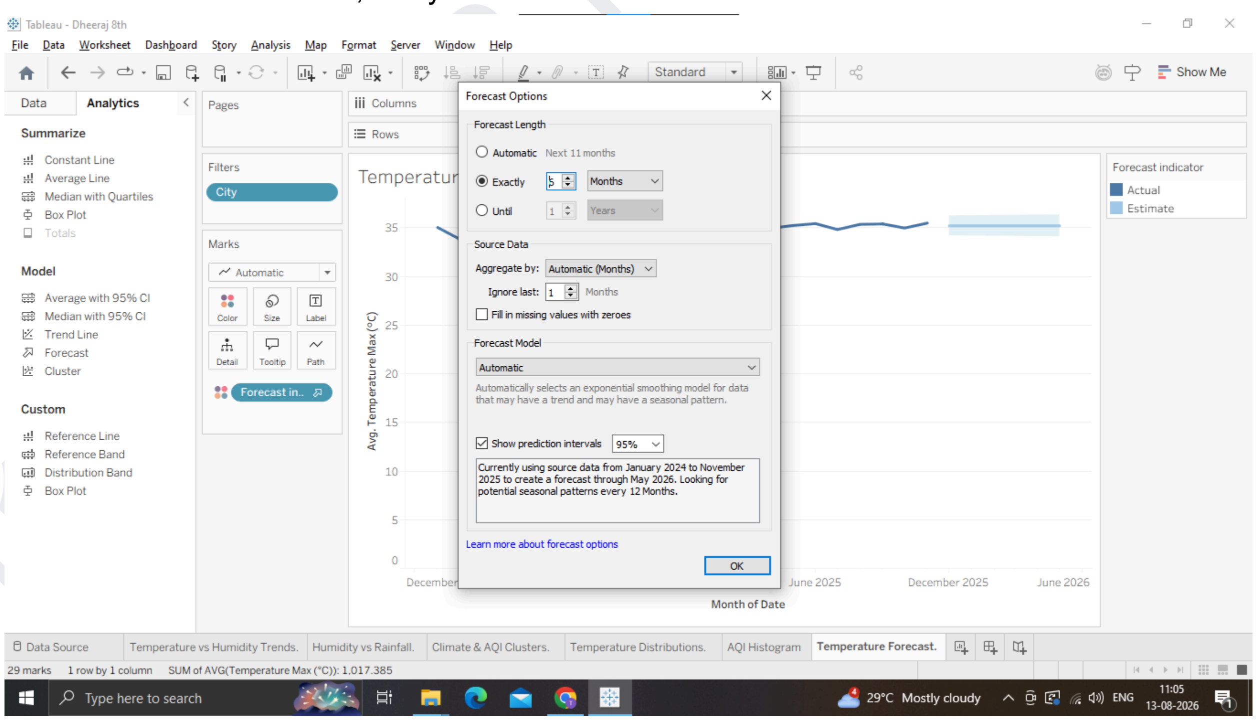
3. Tableau will instantly extend the line chart into the future with a shaded prediction interval band!

3. Customize & Analyze the Forecast

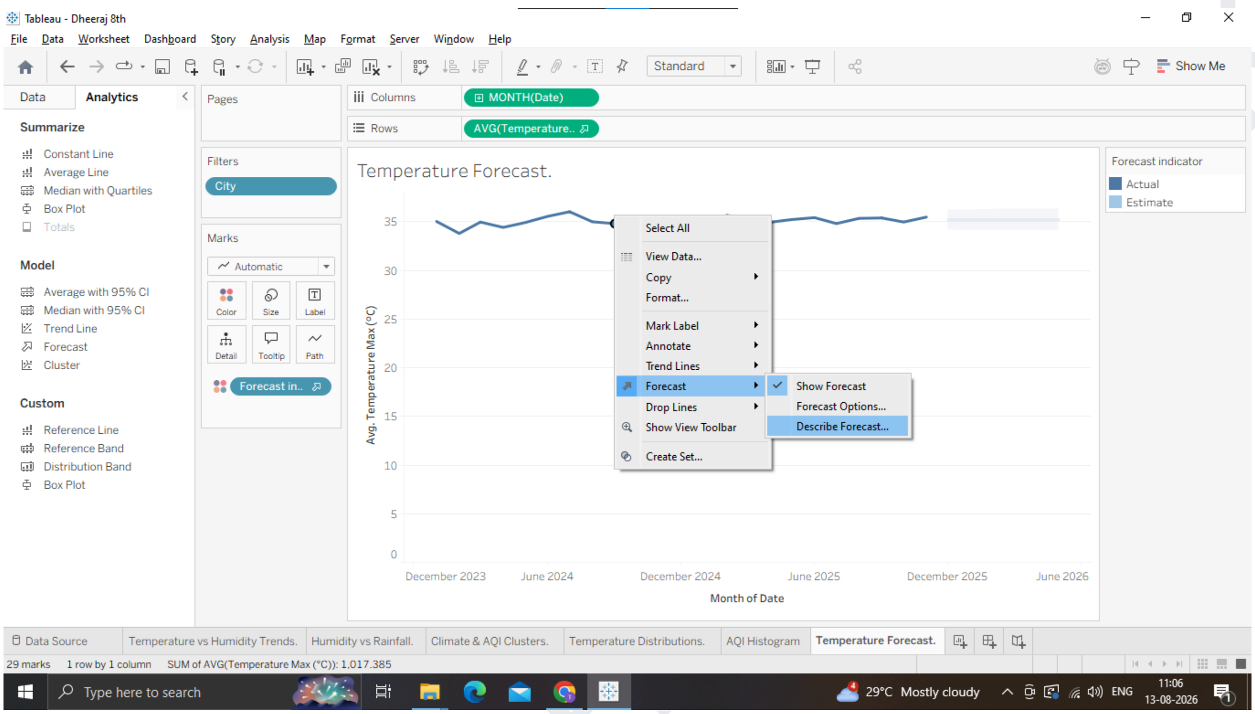
1. Right-click anywhere on the chart canvas select Forecast Forecast Options....

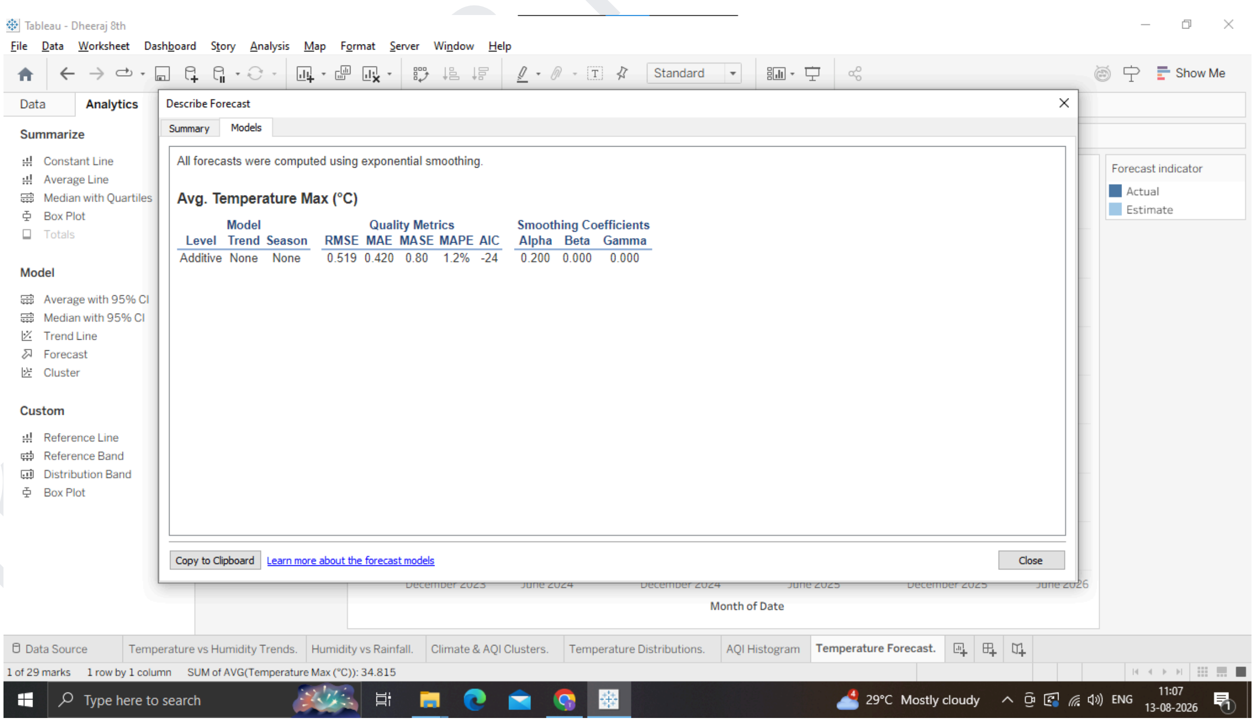


2. Under Forecast Length, set it to Exactly 6 Months into the future.
3. Under Prediction Interval, verify it is set to 95%.



4. To evaluate the mathematical quality of the forecast:
 o Right-click the chart - Forecast Describe Forecast....
 o Review the Models tab to check metrics like the Root Mean Squared Error (RMSE) and whether Tableau selected an Additive or Multiplicative model for trend and seasonality.





Model			Quality Metrics					Smoothing Coefficients		
Level	Trend	Season	RMSE	MAE	MASE	MAPE	AIC	Alpha	Beta	Gamma
Additive	None	None	0.519	0.420	0.80	1.2%	-24	0.200	0.000	0.000

Why we are doing it ?

Predict Future Trends: Forecasting extends historical time-series data into the future using statistical modelling (specifically, exponential smoothing).

Seasonality & Trend Detection: Tableau's forecasting tool automatically evaluates underlying patterns, such as seasonal spikes (e.g., summer temperature peaks or monsoon rainfall) and long-term climate drift.

Quantify Uncertainty: Forecasts include Prediction Intervals (e.g., 95% confidence bands) to show the range within which future values are statistically likely to fall.

Usage

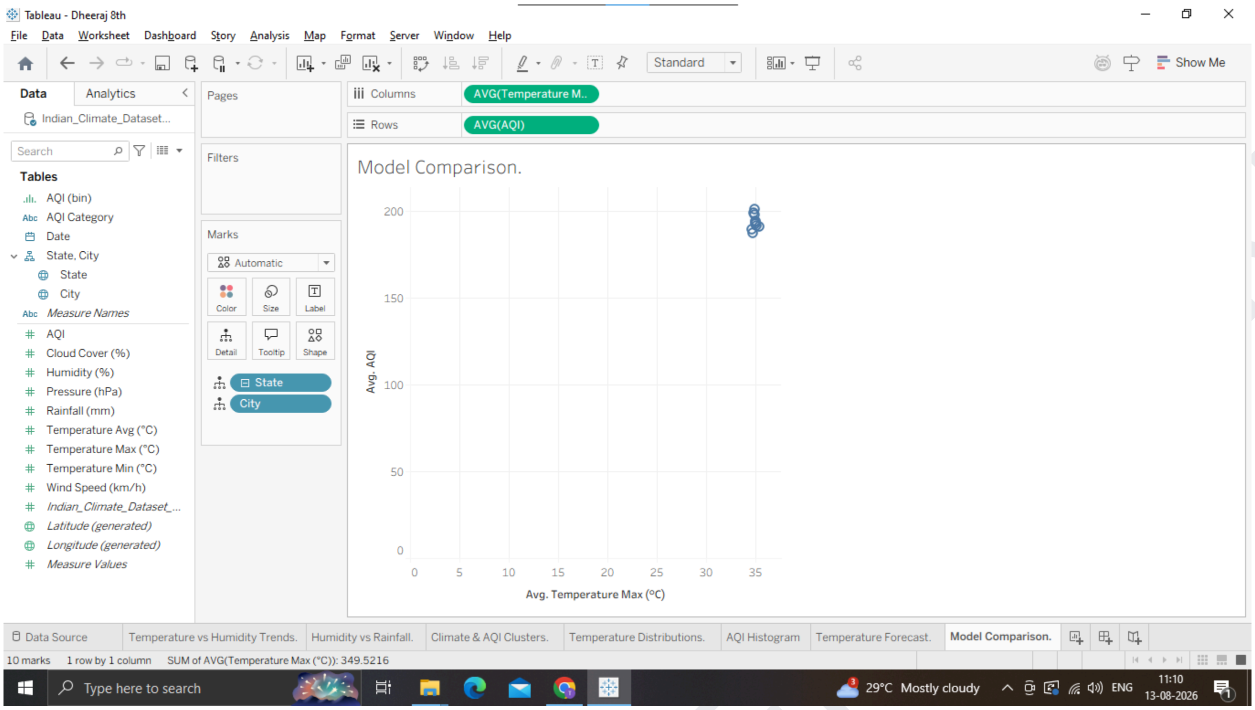
Energy Demand Planning: Power grid operators use temperature forecasts to anticipate peak air-conditioning or heating loads months in advance.

Agriculture & Water Resource Management: Forecasting seasonal rainfall trends helps local governments manage reservoir levels and advise farmers on crop cycles.

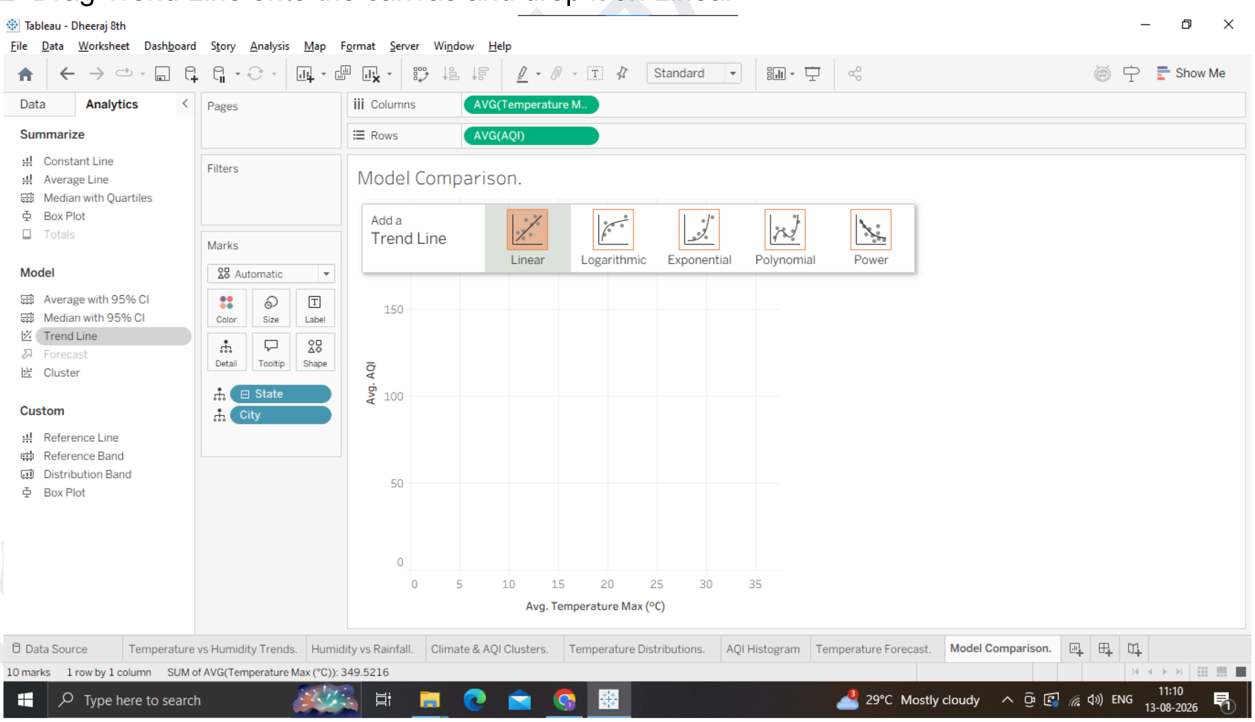
F: Compare different Statistical Models

1. Set Up the Comparison View

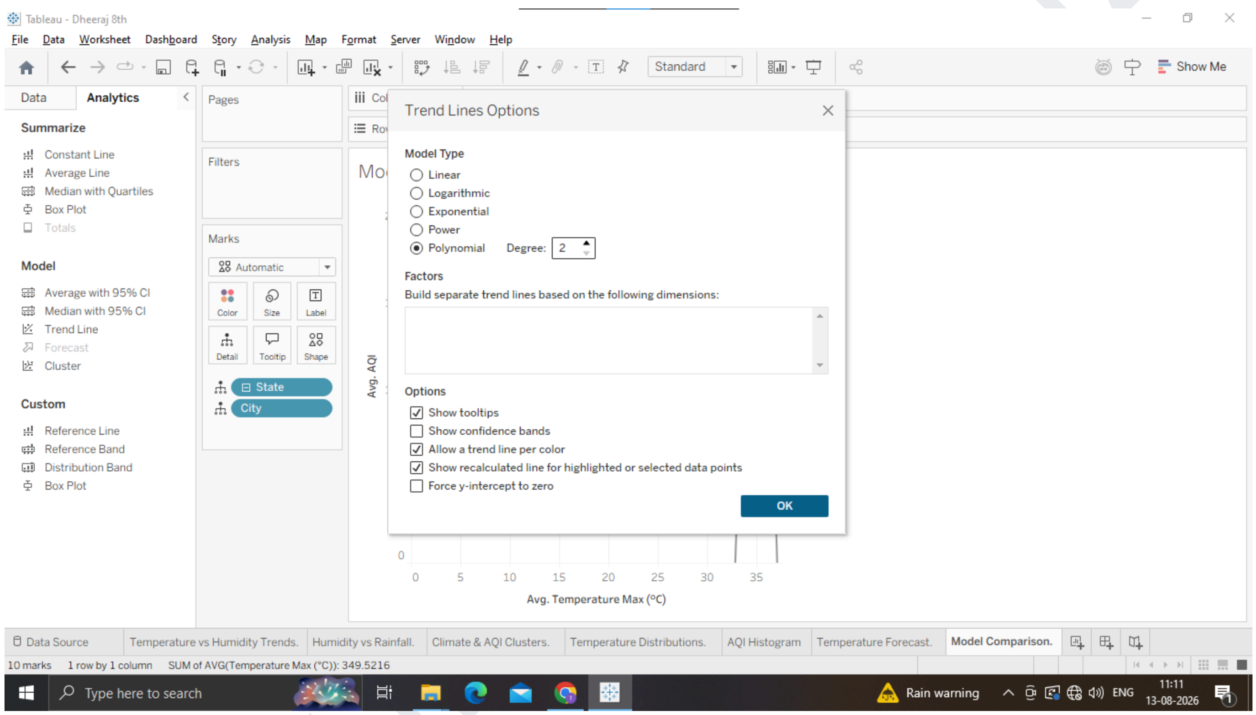
1. Open a new worksheet named Model Comparison.
2. Drag Temperature_Max (°C) to Columns (set to Continuous Dimension or AVG).
3. Drag AQI to Rows (set to Continuous Dimension or AVG).
4. Drag City to Detail on the Marks card to scatter plot individual city readings.



2. Add and Swap Statistical Models
1. Open the Analytics Pane on the left.
2. Drag Trend Line onto the canvas and drop it on Linear.



3. Hover over the trend line and note the R2 and p-value.
4. Right-click the trend line Edit Trend Lines....
5. Switch the Model Type through the options:
o Linear
o Logarithmic
o Exponential
o Polynomial (Degree 2 or 3): Fits curved relationships.
6. Compare the R2 value for each model type. The model with the higher R2 and a p-value < 0.05 provides the best fit.



Why we are doing it ?

Model Validation: Not all trends are linear. Comparing statistical models (e.g., Linear, Exponential, Polynomial, Power) helps identify which mathematical function best describes the underlying data.Goodness-of-Fit: We evaluate models using key metrics like R2 (Coefficient of Determination) which measures how much variance is explained by the model and the pvalue which tests statistical significance.

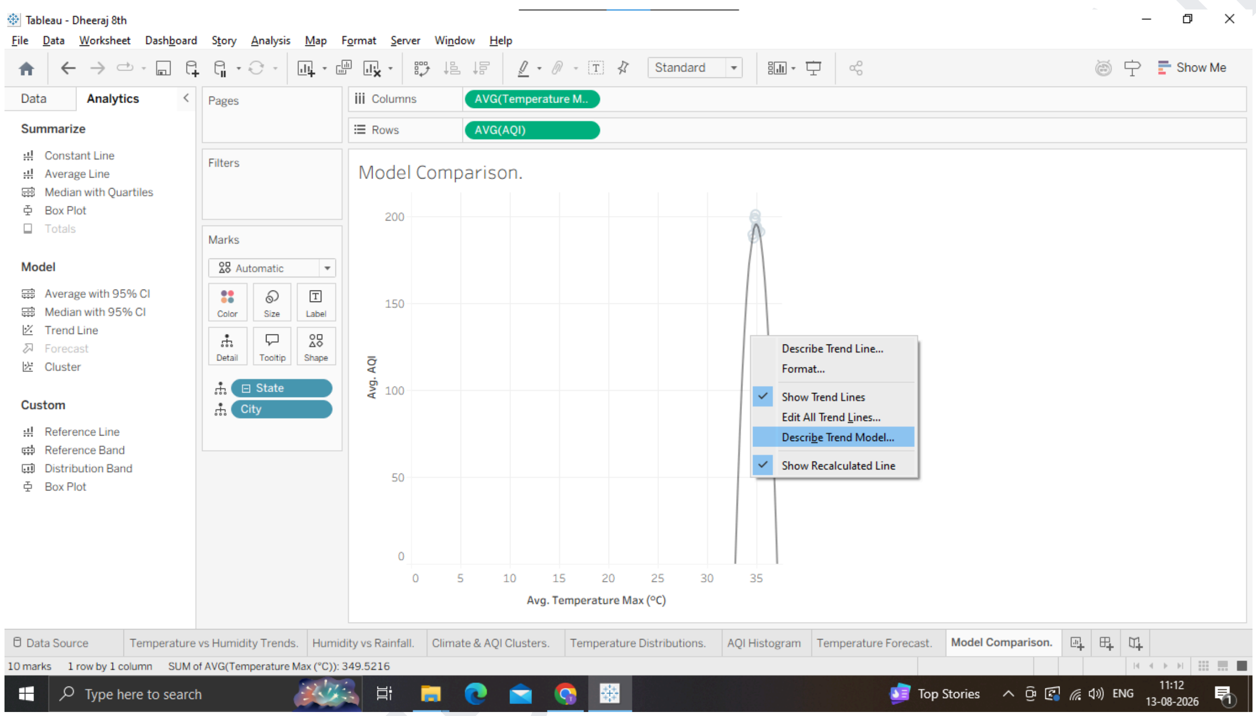
Usage

Selecting Climate Models: Temperature growth might be linear over short periods, but pollutant concentration or population impacts often follow exponential or polynomial curves. Selecting the right model prevents underestimating future risks.

G: Export and Interpret Statistical Results

1. Export the Statistical Model Description

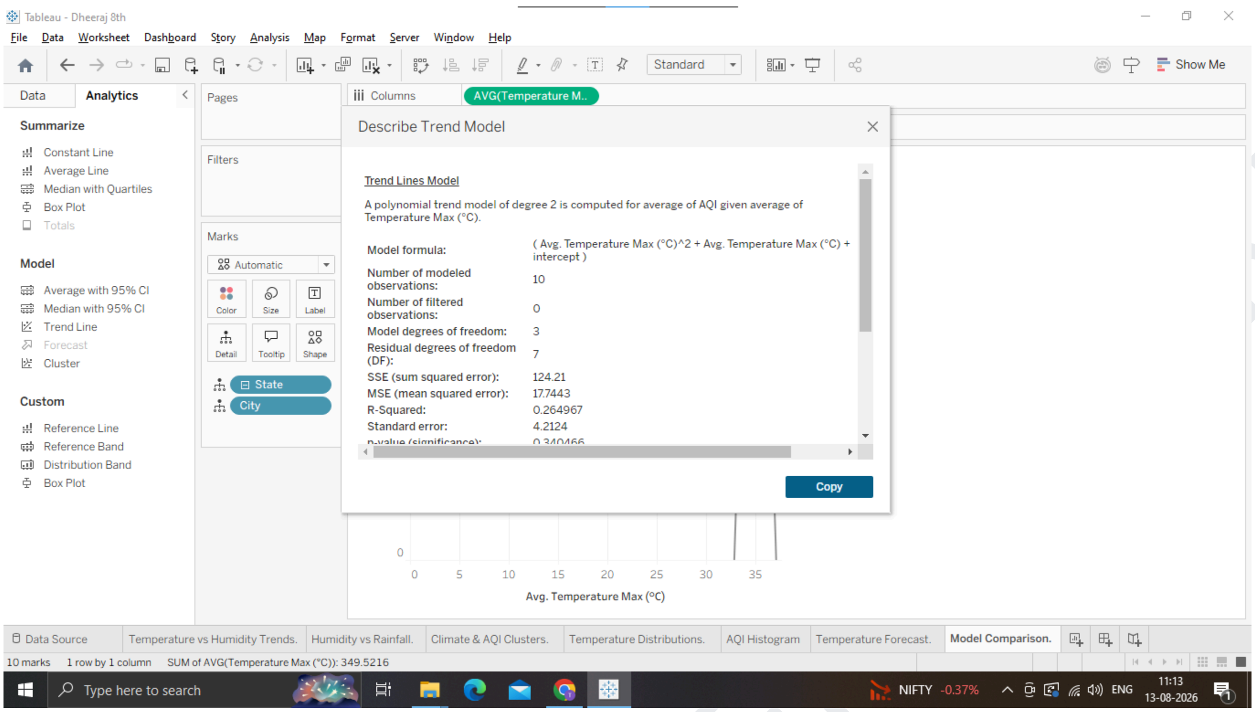
1. On your Model Comparison worksheet (or any trend/forecast sheet), right-click the trend line or canvas.
2. Select Describe Trend Model... (or Describe Forecast... if on a forecast sheet)



3. In the pop-up window, click Copy at the bottom (or click Export to save as a text file).
4. You now have the full ANOVA summary table and parameter estimates copied to your Clipboard!

2. Interpret the Key Output Metrics When reviewing or documenting your exported statistics, highlight these 3 core findings:

1. Equation Parameters: Look at the formula coefficients (intercept and slope).
2. R2 Value:
o Example: An $R^2 = 0.74$ means 74% of the variance in AQI is explained by Temperature changes.
3. p-Value: o A p-value < 0.0001 indicates the relationship is statistically significant (unlikely to occur by random chance).



Why we are doing it

Reporting & Auditing: Data science and analytics workflows require sharing the underlying statistical summary (coefficients, degrees of freedom, residuals) with external teams or academic reports.

Formal Interpretation: Translating Tableau's automated outputs into actionable narrative insights.